

Graph Learning and Node Subset Selection under Sparsity Constraints

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Chapter 1

Introduction

1.1 Subset Selection Problem

Selection of the *right* subset, traditionally enjoys the legacy of being one of the most sought-after problem. Thanks to the ever-expanding world, which makes this problem never lose its charm. Choosing the best representative from a large domain is a problem that fundamentally being ubiquitous and NP-Hard at the same time, suggests always in all contexts where this problem arises, that there is a pressing need for optimizing the procedure of selecting the right subset. The problem has seen many applications to it occurring almost every possible walks of life, be it from picking up the right subset of points to pick from the set of points, to appoint as the cluster heads, and then dispersing off the rest of the points to those zonal heads, which could be commonly called as clustering or picking up the right representative subset from the population which will help identify the full population dynamics. All in all this problem of Subset selection emphasizes strongly an optimization problem, which wants to maximize a function from the minimum possible observatory or scientific effort.

CHOOSING THE CLUSTERING “HEADS” TO BE THE REPRESENTATIVES

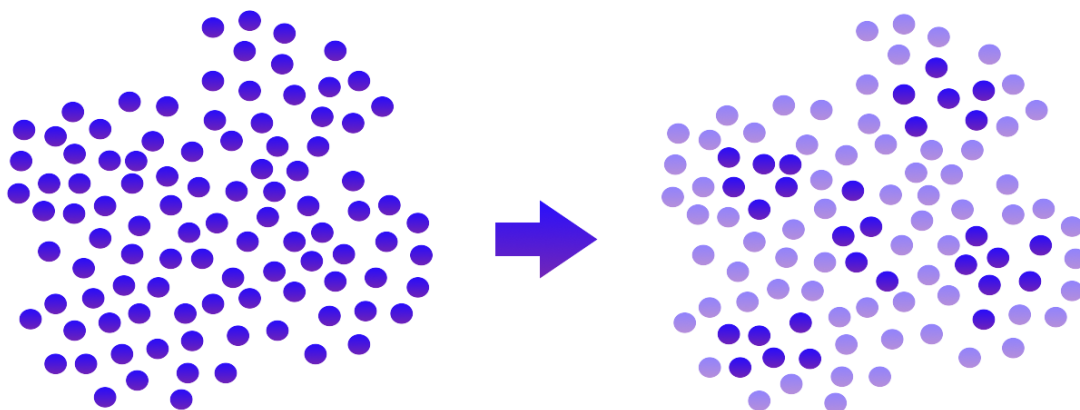


Figure 1.1: Picking Up the “Right” Subset

Presented here is yet another application of Subset selection in the context of Speedy group testing on the verge of an Epidemic. An interesting primer on epidemics and epidemiology is presented below: An epidemic is like a sudden storm

of sickness that sweeps through a community, leaving a trail of coughs, fevers, and aches in its wake. It's like a wildfire of infection, spreading rapidly and threatening to overwhelm the healthcare system. It's a wake-up call that reminds us of the power of infectious diseases and the importance of public health measures to keep them at bay. Epidemiology is like a detective story, but instead of solving crimes, the goal is to unravel the mysteries of diseases and outbreaks. It's a field of study that combines the power of science and data to understand how diseases spread, who is most vulnerable, and how to prevent and control them. Epidemiologists are like disease detectives, using their skills to track down clues and patterns, piece together evidence, and make sense of complex health issues. With epidemiology on our side, we can uncover the secrets of diseases and protect our communities from their deadly grip.

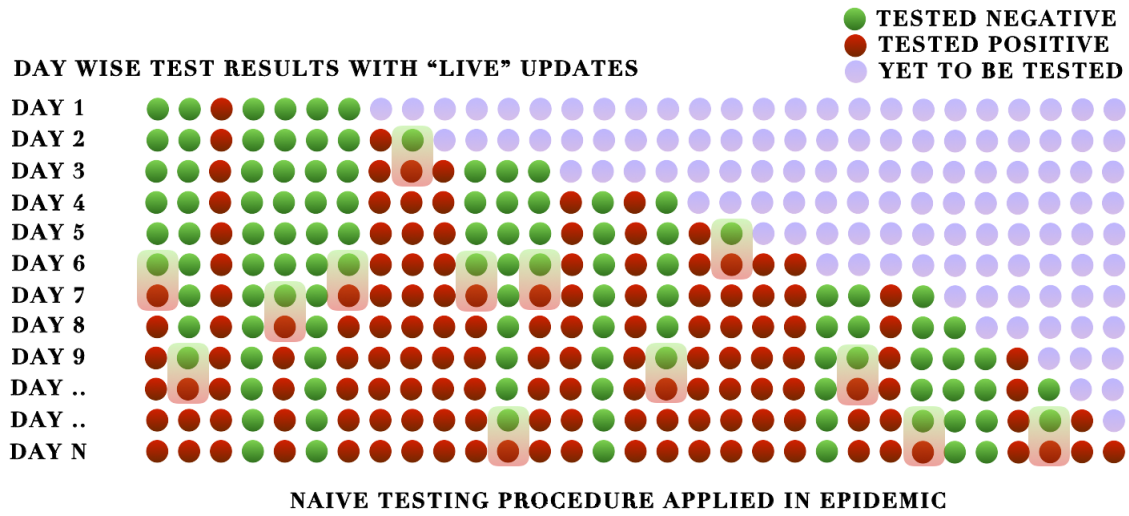


Figure 1.2: The *Testing* “PARADOX”

1.2 Dynamics of Epidemiology in an Epidemic

In the entire Epidemiological narrative, it's illogical to proceed about almost anything unless we know exactly who has caught up the disease and who hasn't. So it's quite succinct that one of the most vital element to begin any sort of epidemiological modeling is “*Testing*”. Obviously, the naive approach is to begin testing every single individual in the region and then tinker with ways to curb the viral outbreak. But the irony is that by the time we reach to test the 800th person we might find the 1st person to whom when we tested earlier in the beginning and tested negative, has been caught with the disease. So all efforts got wasted? Should we restart the entire testing campaign? or Should we just not care about all this and take action based on whatever data is collected? The need of the hour is not to answer these questions but to thoroughly understand those questions, to understand exactly where all this is going, and understand the subtleties lying underneath the core problem. All the above was just regarding one such typical problem that might be encountered while going about Testing, there could be tons of other problems waiting to be questioned and solved. A principled approach to testing is paramount and should be regarded as the highest priority task at hand, in the World of epidemiology.

1.3 BMC - Measles Dataset

Work done over “Testing”, presented herein, is in the Measles Disease spread context, for which the Dataset comes from the Brihanmumbai Municipal Corporation “BMC”, below is a complete breakdown of all the possibly relevant information that could be deduced from the corpus.

Age Bin	Average	Min	Max	Count
0-1	0.63	0	0.92	349
1-2	1.38	1	1.92	238
2-3	2.37	2	2.92	131
3-4	3.42	3	3.92	88
4-5	4.43	4	4.92	90
5-6	5.33	5	5.92	64
6-7	6.42	6	6.92	53
7-8	7.35	7	7.92	62
8-9	8.4	8	8.83	37
9-10	9.41	9	9.92	33
10-12	10.81	10	11.92	51
12-16	13.47	12	15.67	30
16-20	17.93	16.08	19.92	20
20-25	21.93	20	24.75	18
25-30	28.17	25.67	29.92	8
30-40	32.125	30.25	35.75	6
40-50	42.73	40.42	47.33	5
50-60	50.63	50	51.25	2
60-70	62.5	60	65	2

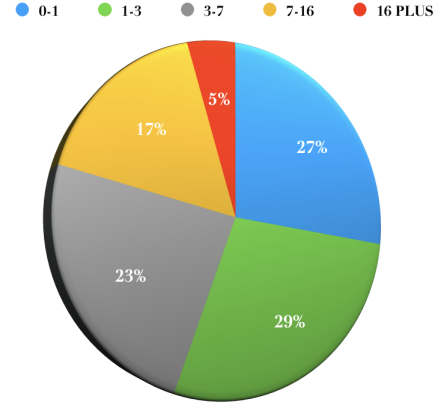


Figure 1.3: Age Wise Statistics

The Pie chart along with the summarized dataset shows the age-wise breakdown of the data inferred from the BMC-provided dataset. It shows there is a huge dominance of infants rather than adults!

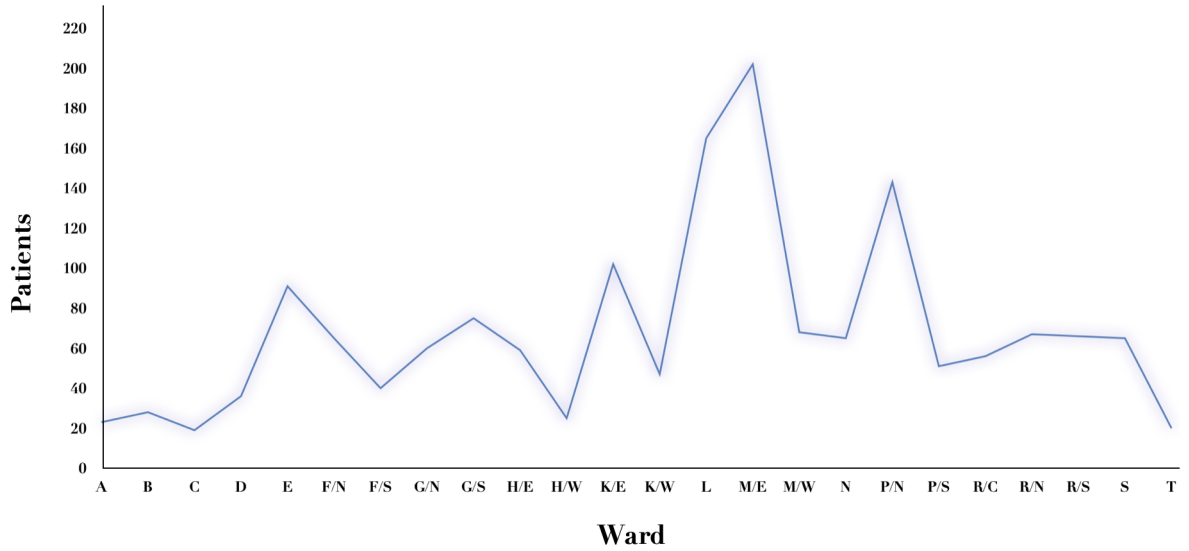


Figure 1.4: Ward Wise Statistics

The Line-Graph shows the Ward-Wise Statistics inferred from the BMC dataset. The above graph lists the concentration of the number of patients across

the wards in Mumbai.

The Skewness of the Graph, quite succinctly speaks about the higher concentration of infection spread in various regions. Wards listed on the X-axis are actually consecutively located in Mumbai, thus very elegantly one can infer the pattern of the Infection Spread for adjacent wards across Mumbai City by a mere stare at it.

A very nice “Bell-Shaped” Curve is formed here, indicating the higher concentrations of Infection near Ward L and M/E and thereby the infection intensity just feels like smearing away across wards on either side.

Hospital	Patients
Kasturba Hospital	190
J J Hospital	26
Jijamata Nagar Dispensary	19
Jijamata Health Post	15
Rajawadi Hospital	14
Cooper Hospital	12
Dr. B.R. Ambedkar Hospital	12
Dr. Prashant Gandhi Hospital	12
Dr. Rupesh Mandodkar Hospital	12
Marol Health Post	12
Malvani Health Post	11
D.R. Shinde’s Children Hospital	10
Gov General Hospital	10
M.W. Desai Hospital	10
Mankhurd Health Post	10
Maruti Lane Dispensary	10
Subhash Nagar Health Post	10
Others	892

Figure 1.5: Hospital Wise Statistics

The Table shows the Hospital-wise Statistics inferred from the BMC dataset. The above list shows the number of people who were reported to be infected by each hospital. Listed here are only the Hospitals that had at least 10 people reported as Infected and all other hospitals had less than 10 reported cases and hence categorized as “Others” above.

Graphs seen till now were among the most important aspects one could infer from the dataset, although there are some more modalities that need to be explored from the dataset, like, Pin Code Wise Statistics, Travel Frequency via Public Transports viz. Railways, the BEST Busses, etc, and because the infected population is predominantly dominated by infants and school-going children, a track over the school location could also lead us to deeper insights related to what the actual infection spread hubs are.

Chapter 2

Graph Signal Processing

In this Chapter we will talk in general about Graph Signal Processing and Graph Laplacian.

2.1 Graph Signal Processing

Graph signal processing (GSP) is a field that deals with signals defined on graphs, where graphs are mathematical objects consisting of vertices and edges that model relationships between different entities. In GSP, signals are represented by vectors whose components correspond to values assigned to vertices in a graph.

One of the key challenges in GSP is to develop methods that can handle the high-dimensional nature of signals on graphs. Since graphs can have a large number of vertices and edges, signal processing techniques that are designed for low-dimensional data may not be directly applicable to graphs.

To overcome this challenge, GSP researchers have developed a variety of techniques that leverage the underlying graph structure. For example, the graph Laplacian matrix, which encodes information about the relationships between vertices, has been used to define graph Fourier transforms and associated frequency-domain representations of graph signals. This has allowed researchers to apply tools from classical Fourier analysis to the graph setting, enabling them to develop new filtering, sampling, and compression methods for graph signals.

Another key area of research in GSP is the development of graph learning algorithms, which aim to learn the underlying graph structure from data. This is an important task in many applications, such as social network analysis, where the graph structure may not be known a priori.

Overall, GSP is an exciting and rapidly growing field that has the potential to revolutionize signal processing by providing powerful tools for analyzing and processing signals on graphs.

2.2 Graph Spectral Domain

Spectral graph theory has traditionally focused on the construction, analysis, and manipulation of graphs, rather than signals on graphs. It has proven to be particularly useful in the creation of expander graphs, graph visualization, spectral

clustering, graph coloring , and many other applications in chemistry, physics, and computer science.

In the field of signal processing on graphs, spectral graph theory has been utilized as a tool to define frequency spectra and expansion bases for graph Fourier transforms. In this section, we will provide a brief overview of some fundamental definitions and notations from spectral graph theory, emphasizing how it enables us to extend many of the important mathematical concepts and intuitions from classical Fourier analysis to the graph setting.

2.3 Graph Laplacian

The unnormalized graph Laplacian, also known as the combinatorial graph Laplacian, is defined as:

$$L = D - W \quad (2.1)$$

where D is a diagonal matrix with the i th diagonal element d_i equal to the sum of the weights of all the edges incident to vertex i , and W is the adjacency matrix of the graph.

The Laplacian is a difference operator for any signal f in R^N , and it satisfies:

$$(Lf)(i) = \sum_{j \in N_i} (W_{i,j}[f(i) - f(j)]) \quad (2.2)$$

where N_i is the set of vertices connected to vertex i by an edge, and $N(i, k)$ is the set of vertices connected to vertex i by a path of k or fewer edges.

The graph Laplacian L is a real symmetric matrix with a complete set of orthonormal eigenvectors denoted as $u_0, u_1, \dots, u_{N-1}$. The associated eigenvalues are real and nonnegative, denoted as $\lambda_0, \lambda_1, \dots, \lambda_{N-1}$, satisfying $Lu_i = \lambda_i u_i$ for $i = 0, 1, \dots, (N-1)$. The eigenvalues are ordered as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$. The number of connected components of the graph determines the multiplicity of the eigenvalue 0, and since we consider connected graphs, we assume that the graph Laplacian eigenvalues are ordered as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$. The set of all eigenvalues is denoted as $v(L) = \lambda_0, \lambda_1, \dots, \lambda_{N-1}$.

2.4 Compressed Sensing Theory for Reconstruction of Partially known Signals

Compressed sensing (CS) is a powerful framework for reconstructing unknown signals from sparse measurements. It has been widely used in various signal processing applications, including image processing, signal acquisition, and data compression. In the context of graph signal processing, CS has been applied to reconstruct graph signals from a limited number of measurements.

The basic idea of CS is to leverage sparsity, or compressibility, of signals in some basis or transform domain. The key observation is that many real-world signals can be represented by a sparse or compressible linear combination of basis functions. In other words, most of the signal energy is concentrated in a few basis coefficients, while the rest of the coefficients are close to zero. CS exploits this

sparsity property by acquiring a small number of linear measurements of the signal and then using an appropriate reconstruction algorithm to recover the original signal from the measurements.

In the graph signal processing setting, the graph Laplacian matrix plays a central role in CS-based signal reconstruction. The graph Laplacian encodes the structural information about the graph, including the connectivity and the degree of each vertex. It also provides a natural framework for defining a graph Fourier transform and associated frequency-domain representations of graph signals.

To apply CS to graph signals, one can represent the graph signal in the graph Fourier domain, where the signal is sparse or compressible. The sparse representation can be obtained by solving an optimization problem that minimizes the ℓ_1 -norm of the graph signal in the graph Fourier domain subject to the constraint that the measurements are consistent with the acquired data. The resulting optimization problem is a convex optimization problem that can be efficiently solved using various algorithms such as iterative soft thresholding (IST) and orthogonal matching pursuit (OMP).

The key challenge in applying CS to graph signals is to design appropriate measurement matrices that allow for accurate signal reconstruction from a small number of measurements. In general, the measurement matrix should be incoherent with the graph Fourier basis and should satisfy certain properties such as the restricted isometry property (RIP) to ensure accurate signal recovery.

In summary, compressed sensing theory has been applied to graph signal processing by leveraging the sparsity property of graph signals in the graph Fourier domain. The graph Laplacian matrix provides a natural framework for defining the graph Fourier transform and associated frequency-domain representations of graph signals. CS-based signal reconstruction can be achieved by solving a convex optimization problem that minimizes the ℓ_1 -norm of the graph signal in the graph Fourier domain subject to the constraint that the measurements are consistent with the acquired data. Appropriate measurement matrices that are incoherent with the graph Fourier basis and satisfy certain properties such as RIP are essential for accurate signal recovery.

Chapter 3

Graph Signal Processing for Testing

In this chapter we will deeply analyse 2 research papers which have worked in this field, viz., "Graph Frequency Analysis of Covid-19 Incidence to Identify County level Contagion Patterns In the United States" by Li et. al. and "Kemeny-based testing for COVID-19" by Yilmaz et. al.

3.1 Graph Frequency Analysis of Covid-19 Incidence to Identify County level Contagion Patterns In the United States

The study examines the spread patterns of COVID-19 in US counties through a graph frequency analysis. Commute flows between all 3142 US counties were used to construct a graph representing population mobility, and the numbers of daily confirmed cases were collected as graph signals. The graph Fourier transform was used to map the graph signals into the frequency domain, enabling the decomposition of signals into modes with smooth or rapid variations. The study shows that different modes of variability relate to COVID-19 spread patterns within and across counties, revealing changes in the nature of spread within geographical regions at finer temporal scales. The approach demonstrates the potential of using Graph Signal Processing (GSP) tools for epidemiology knowledge discovery on graphs.

Graph representations of networks enable the analysis of complex data from various aspects of human life. Graph Signal Processing (GSP) emerged to leverage graph structures for information processing, and it has been applied to various timely applications. This study uses recent GSP advances to investigate the spread pattern of COVID-19 across all US counties, providing a comprehensive spatio-temporal analysis of the contagion. The study involves spatio-temporal analysis of case data from 3142 US counties, graph frequency analysis using a GFT-based approach, and investigation of changes in the nature of contagion patterns within counties. The results reveal fundamentally different contagion patterns across locations and help identify counties at risk of major outbreaks in a data-driven manner.

3.1.1 GSP fundamentals

The Paper considers signals that are associated with nodes of a weighted, undirected, and connected graph. More formally, we can define the signal vector $x = [x_1, \dots, x_N]^T \in R^N$ and the weighted graph $G(V, E, W)$, where $V = 1, \dots, N$ represents the set of nodes and $E \subseteq V \times V$ is the set of edges. Here, x_i denotes the signal value at node $i \in V$, and the map $W : V \times V \rightarrow R^+$ associates a non-negative weight $W_{ij} \geq 0$ with the edge $(i, j) \in E$, while $W_{ij} \equiv 0$ for $(i, j) \notin E$. The symmetric coefficients $W_{ij} = W_{ji}$ capture the strength of the connection between nodes i and j , reflecting their similarity or influence.

For the purpose of this paper, they focus on a specific graph consisting of $N = 3142$ counties in the United States. The edge weights in this graph correspond to the average amount of population commuting between counties. The signal x represents the daily number of confirmed COVID-19 cases across all counties. One can interpret the edge weights as capturing the strength of the connection between counties, which reflects the potential for virus transmission based on population commuting flows. If the weight W_{ij} is large, the signal values x_i and x_j tend to be similar, indicating a higher likelihood of virus transmission. Conversely, if the weight W_{ij} is small, the signal values x_i and x_j are less directly related, except through their weak connections to other nodes.

This interpretation of the edge weights establishes a link between the signal values and the graph topology, motivating the use of a novel Graph Signal Processing (GSP) approach, which is proposed in this paper.

3.1.2 GFT and Smoothness

GFT which is Graph Fourier transform is an important tool in the Graph Signal Processing World, as it allows to represent the graph signal into 2 different domains:

1. Vertex Domain, consisting of the nodes in V
2. Graph Frequency Domain spanned by the spectral basis of G , Thus to induce the different levels of interaction between the neighbors the signals can be manipulated in the frequency domain.

To expound on this concept, we can use the eigenvector decomposition of the combinatorial graph Laplacian $L := \text{diag}(W_1) - W$ to define the Graph Fourier Transform (GFT) and the associated concept of graph frequencies. We can always decompose the symmetric graph Laplacian as $L = V\Lambda V^T$, where $\Lambda := \text{diag}(\lambda_1, \dots, \lambda_N)$ denotes the diagonal matrix of non-negative Laplacian eigenvalues and $V := [v_1, \dots, v_N]$ is the orthonormal matrix of eigenvectors.

The GFT of a signal x with respect to the Laplacian L is the signal $x_\sim = [x_\sim 1, \dots, x_\sim N]^T$ defined as $x_\sim = V^T x$. The inverse GFT (iGFT) of $x_\sim$ is given by $x = V x_\sim$, which is a proper inverse due to the orthonormality of V .

The formula for iGFT, $x = V x_\sim = \sum_{k=1}^N x_\sim^{(k)} v_k$, allows for the synthesis of x as a sum of orthogonal frequency components v_k , where the contribution of each v_k to the signal x is given by its corresponding GFT coefficient $x_\sim^{(k)}$. The GFT captures the concept of signal variability over the graph, similar to how frequency captures variability in Fourier analysis of temporal signals.

To understand this analogy, we can define the total variation of the graph signal x with respect to the Laplacian L (also known as Dirichlet energy) as the

quadratic form $TV(x) := x^\top Lx = \sum_{i \neq j} W_{ij}(x_i - x_j)^2$. The total variation $TV(x)$ serves as a measure of smoothness, quantifying the amount of change in the signal x with respect to the variability encoded by the weights W .

Returning to the GFT, we can examine the total variation of the eigenvectors v_k , which is given by $TV(v_k) = v_k^\top Lv_k = \lambda_k$. Therefore, the eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$ can be interpreted as graph frequencies, indicating how the eigenvectors (i.e., frequency components) vary over the graph G . As a result, the GFT and iGFT provide a means of decomposing the graph signal x into spectral components that capture different levels of variability.

For a graph signal x with GFT coefficients x , filtering can be performed in the frequency domain similarly to classical signal processing of time-varying signals. Eigenvalues of the Laplacian correspond to graph frequencies and eigenvectors act as a frequency basis. Using this, low-pass, band-pass, and high-pass graph filters can be defined by isolating specific eigenvalues and their corresponding eigenvectors. These filters are mutually exclusive and span all graph frequencies. By mapping signals to the spectral domain via the GFT, filtering them, and then mapping each frequency component back to the vertex domain via the iGFT, the original graph signal can be decomposed into components exhibiting low, medium, and high variability with respect to the graph. This decomposition increases the resolution of the signal and can be used to investigate different patterns across the graph.

3.1.3 GSP and COVID-19

Graph frequency analysis conducted on US COVID-19 prevalence data, defining signals and constructed graph for study, followed by thorough frequency analysis to identify contagion patterns across counties.

COVID-19 data as Graph Signals

The section discusses how COVID-19 data can be represented as graph signals by using the number of daily confirmed cases per 100k residents of each county in the US, and stacking these signals as columns of a matrix. The study takes into account the commute flow between counties in building a network graph, which helps to reveal hidden relationships between signals and identify different patterns of virus contagion across different locations in the country.

Network Graph Construction

The study uses data from the US Census Bureau to construct a weighted undirected graph with 3142 counties as nodes and edges representing population mobility flows between counties, normalized by intra-flows. The resulting graph shows dense diagonal blocks as expected due to mobility within states, but also significant connections between counties in different states, suggesting long distance commutes. This network provides richer pairwise similarity patterns than the distance-based graph used in previous work.

Frequency decomposition of graph signals

In this section, the authors use the graph signals X and graph G from the

previous sections to carry out frequency decomposition of the daily case signals. They form the Laplacian matrix using W and compute its eigenvalues and eigenvectors. A low/high-pass filter is constructed by taking the lowest/highest one fifth of the eigenvalues, respectively. After graph filtering, the original graph signal X is decomposed into XL, XM, XH which represents signal components that change slowly/mildly/rapidly with respect to the underlying commute graph G . The authors focus on low-pass and high-pass signals as they correlate with distinct contagion patterns. They take the row-wise average of the absolute values in XL and XH to obtain two vectors of length 3142 that quantify the signal magnitude per county with respect to their energy occupancy in low-pass and high-pass spectral band. They use LP regions to represent counties with high magnitude of low-pass signals, and HP regions for counties with high magnitude of high-pass signals. They decompose the signal of each county into low-pass and high-pass frequency contributions and assess whether its signal concentrates more in either frequency band. They overlay the magnitudes of low-pass signals and high-pass signals of all counties on a US map and observe that LP regions have similar number of daily confirmed cases normalized by population size with neighboring counties, while HP regions have markedly distinct and abnormal case counts relative to nearby counties, despite the population mobility between them. Analysis in the next section establishes a link between graph frequencies (low/high-pass) and contagion spread patterns based on descriptive analysis of the graph signals.

Frequency Analysis

They analyze the frequency components of the GFT to identify patterns of spread across counties. The authors explain that low-pass signals (smooth, diffusive signals) in the GFT correspond to cross-county contagion patterns, where nearby counties tend to exhibit similar numbers of confirmed cases. On the other hand, high-pass signals (spiky signals) correspond to within-county outbreaks, where the signal of a county is very dissimilar to nearby counties. The authors use these frequency components to identify regions with cross-county spread and within-county outbreaks.

The authors compute several metrics for these regions, such as average daily cases and the number of days to reach peaks. They find that LP regions have more daily cases and take longer to reach peaks, while HP regions have fewer daily cases and reach peaks faster. They argue that LP regions suffer from cross-county spread with a continuous increasing rate and a longer contagion period, while HP regions suffer from local outbreaks in highly concentrated places.

Finally, the authors analyze the temporal evolution of COVID-19 cases in the New York City area, using weekly periods. They show how a localized outbreak in one county transforms into a wider spread across the region, as seen in the dominance of low-pass signals in the GFT. The authors argue that their approach provides insights into the spread of COVID-19 across different regions and can inform public health policies.

3.2 Kemeny-based testing for COVID-19

At its core, Authors are discussing the importance of testing, tracking, and tracing in controlling the spread of COVID-19 and safely reopening activities. Contact tracing apps allow for the reconstruction of daily contact graphs, which can be used to identify bridges between different communities. The authors propose a new metric called the "Kemeny indicator," which is based on the Kemeny constant of a graph and can be used to identify individuals who are likely to be "super-spreaders" or links between different communities. By testing individuals associated with high Kemeny indicators, early identification and interception of new virus outbreaks can occur. The effectiveness of this approach has been demonstrated through extensive simulations.

Kemeney indicator for criticality

Authors describes a method for identifying the most critical nodes in a graph, which are those that connect different communities and could potentially act as "super-spreaders" in the transmission of disease. This method involves calculating the Kemeny constant of the graph after removing each node and using this value as a measure of the node's importance. Nodes with high Kemeny indicators are those that, if removed, would cause a significant increase in walking times between communities, indicating their criticality.

PageRank and Centrality indicator

Here Authors discuss different measures of centrality in graphs, specifically focusing on PageRank and betweenness centrality indicators. It explains how PageRank corresponds to the Perron eigenvector, and that teleportation is a factor in the calculation. The passage also describes the use of Random Walk Betweenness (RWB) as a measure of betweenness centrality, which is shown to be effective in identifying nodes that bridge communities. The Kemeny constant is also introduced as a measure of the average time to take a random walk in the contagion graph, weighted according to likely destinations, representing a compromise between indicators that use only the stationary distribution and those using path-based algorithms. The passage concludes that the Kemeny constant should work in highly connected single-community graphs, as well as in more sparse graphs associated with many unknown sub-communities. The article presents a framework for using the change in the Kemeny constant of a graph to identify and analyse bridges between sub-communities in order to efficiently identify and block new virus outbreaks as early as possible by removing possible 'super-spreader' links that transmit disease between different communities. The Kemeny constant is a new method used in sampling problems that is related to other methods in the literature, such as the random walk betweenness indicator. Advantages of the Kemeny constant include its efficient computation, its ability to take into account weighted graphs and variable transmission probabilities, and its ability to handle directed graphs with asymmetric transmission paths.

Overall Analysis of Paper

This Article describes a simulation study that used the Kemeny indicator to identify "bridge nodes" (individuals who connect different communities) in a network of interactions. In the first simulation, the graph representing interactions was drawn randomly each day, but in the second simulation, the graph had a fixed structure and random interactions were simulated on the edges. The authors suggest that in real-life scenarios, testing or intervening with these bridge nodes proactively before infection (e.g. through quarantine measures or targeted education/enforcement) may be more effective than testing them repeatedly. They also note that there are privacy concerns about centralized government knowledge of social graph structures, and suggest that distributed estimation of the Kemeny constant may be possible using distributed ledger technology. Finally, they mention that their work builds on a previous paper from 2011, in which the Kemeny indicator was used to identify critical nodes and edges in networks. The authors believe that their work can be extended in a number of ways, such as taking into account: dynamic evolution of contact graphs, using optimized combinations of indicators, investigating approximated distributed approaches for computing the Kemeny constant, using more sophisticated epidemiological models, and applying the findings to immunization procedures against other seasonal diseases.

Chapter 4

Graph Signal Processing for Subset Selection with Contact Tracing Graph

4.1 Nomenclature And Algorithm

$\mathcal{N}$	Total Population = $\{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_{\mathcal{N}}\}$ where $\mathcal{P}_i$ represents i^{th} Person in the Population. $\mathcal{N}$ is size of Population set.	Scalar
f	Initially infected Population	Scalar
K	Number of days to Simulate	Scalar
ρ	Regularisation Parameter in Convex optimisation problem.	Scalar
τ	Threshold	Scalar
λ	Parameter used in Probability of i^{th} person getting infected from the j^{th} person	Scalar
ViralLoad	Viral Infection Score assigned to every person.	$\mathcal{N} \times 1$
A	Adjacency Matrix. $A[i][j] = 1$ if i and j came in contact $A[i][j] = 0$ if No contact between i^{th} and j^{th} person.	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Matrix.
T	Time of Duration of Contact between i^{th} and j^{th} person. Random Values assigned.	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Matrix.
S	Signal Strength between i^{th} and j^{th} person. Inversely proportional to the distance between i^{th} and j^{th} person. Random Values assigned.	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Matrix.

Figure 4.1: Nomenclature Part 1

$\mathcal{P}$	$P[i][j] = 1 - e^{-\lambda * T[i][j] * S[i][j] * \max(\text{ViralLoad}[i], \text{ViralLoad}[j])}$ viral load heuristic School, College, Places in Rail Roads, Frequency of transfer of people from that pg	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix,
D	Row wise sum stored at Diagonal Positions. $D[i][j] = \begin{cases} 0 & \text{if } i \neq j \\ \sum_{j=1}^N P[i][j] & \text{otherwise} \end{cases}$	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Diagonal Matrix
$\mathcal{L}$	Laplacian Matrix, $\mathcal{L} = D - \mathcal{P}$	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Positive Semi Definite Matrix.
V	Eigen Vectors of $\mathcal{L}$	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix,
X	Binary vector containing health status People. $X[i]=1$ if infected 0 otherwise	$\mathcal{N} \times 1$ Binary Vector

Figure 4.2: Nomenclature Part 2

θ	Our Conjecture, $X = V * \theta$ [Comes from Compressed Sensing Theory] $\theta = V^{-1} * X$ Because V is Orthonormal set of Eigen vectors, $\theta = V^T * X$	$\mathcal{N} \times 1$ Vector
Y	Subset of X	$m \times 1$ Binary Vector $m < \mathcal{N}$
<i>random_indices</i>	Indices of subset of values of X chosen to be stored in Y. If $Y[i] = X[k] \Rightarrow \text{random_indices}[i]=k$	$m \times 1$ vector
ϕ	Row wise one hot encoded Matrix corresponding to Y. for $i = 1: m$ $\phi[i][\text{random_index}[i]]=1$ otherwise 0	$m \times \mathcal{N}$

Figure 4.3: Nomenclature Part 3

1. Set the Hyper-parameters.

$\mathcal{N}$ The total Population Size.

f Initial Infected set of People.

K Total Number of Days to Simulate.

2. Introducing the Community Structure.

Generate the clique structure. For Example, if the total population is $\{P_1, P_2, P_3, \dots P_{15}\}$ then the cliques generated will be $C_1=\{P_1\dots P_6\}$, $C_2=\{P_5\dots P_9\}$, $C_3=\{P_{10}, P_{11}\}$, $C_4=\{P_{11}\dots P_{15}\}$

I have set the overlap factor between any two consecutive cliques to be 50%

3. Constructing the Initial contact tracing matrix A.

This is basically Adjacency Matrix. Now while filling the adjacency matrix, we will strongly connect individuals of each Clique to every other individual of that clique.

$N = 15$

A Adjacency Matrix

There are 3 cliques are here, $C_1 = \{1..5\}$, $C_2 = \{3..5\}$, $C_3 = \{4...15\}$.

0	1	1	1	1	0	0	0	0	0	0	0	0	0	0
1	0	1	1	1	0	0	0	0	0	0	0	0	0	0
1	1	0	1	1	0	0	0	0	0	0	0	0	0	0
1	1	1	0	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	0	1	1	1	1	1	1	1	1	1	1
0	0	0	1	1	0	1	1	1	1	1	1	1	1	1
0	0	0	1	1	1	0	1	1	1	1	1	1	1	1
0	0	0	1	1	1	1	0	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1	1	0	1	1	1	1	1
0	0	0	1	1	1	1	1	1	1	0	1	1	1	1
0	0	0	1	1	1	1	1	1	1	1	0	1	1	1
0	0	0	1	1	1	1	1	1	1	1	1	0	1	1
0	0	0	1	1	1	1	1	1	1	1	1	1	0	1
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0

Figure 4.4: Algorithm Part 1

4. Setting the Viral Loads and the Infection Dates of all the individuals.

Randomly assigned Infection scores to everybody. Infection dates to initial set of people set to the day 0, and infection dates of every other person set to $K + 1$, indicating they don't have any infection right now.

5. Begin the Simulation Process for K days.

5.1 Generating the Contact Tracing Map A

Infusing some random contacts like wildfire in the Matrix A.

5.2 Constructing the Probability Matrix P using the Time of Duration of Contact , the Signal Strength of Contact and the Viral Load carried by the people who have been in contact.

If $A[i][j] = 1$ Assign a probability score.
--

5.3 Compute the Eigen Vector Matrix of Laplacian Matrix L of P

$V = \text{Eigen_vector}[L[P]]$

The above are going to Orthonormal set of Eigen vectors, because L is going to be symmetric positive semi definite matrix.

5.4 Diffusion Process

Now simulating the spread of infection by utilising the Contact Tracing Map and the Probabilities computed.

5.5 Compute $\theta_0 = V^{-1} * X$

5.6 Construct the Y Vector.

Y is a subset of X of size $m \times 1$ where $m < \mathcal{N}$

5.7 Construct the ϕ Vector.

Row wise one hot encoding of Y vector.

Figure 4.5: Algorithm Part 2

X vector represents the binary health status of all people in the population	..N x 1
theta vector represents the sparse view of X	..N x 1
V encodes the Transition Matrix and the laplacian	..N x N
V*theta represents the predicted status for all people in population	..N x 1
phi is used to select the predicted status for subset selected y	..m x N
phi*V*theta gives status for subset Y	..m x 1

Figure 4.6: Algorithm Part 3

5.8 Solve the Convex optimisation Problem

$\arg \min_{\theta} \ y - \varphi \underline{V\theta}\ _2^2 + \rho \ \theta\ $	
Constraint	$0 \leq V*\theta \leq 1$

5.7 We get θ_1

We compute $X = V * \theta_1$ again to get X_1

This represents the final set of population to select as a subset for testing.

Note: Thresholding needs to be done on X_1 to get Binarized version of it.

So we use the hyper parameter τ .

Figure 4.7: Algorithm Part 4

Chapter 5

Applying Various Subset Selection Techniques

5.0.1 Various Data settings where we apply our Model

We Studied, Compared, and analyzed various techniques in the Node Subset selection for optimal reconstruction, using Compressed Sensing. We tried Applying our Model to various data regimes, The Measles Dataset procured from BMC, Measles dataset procured from WHO, American Covid 19 dataset procured from JHU.

TRACKS

BMC MEASLES

- **Node Granularity 533 Nodes** ¹⁰
- **Hospital Independent 407 Nodes** ⁵
- **Village Independent 126 Nodes** ⁶

BMC MEASLES WHO

- **Node Granularity 310 Nodes** ⁷
- **Ward Granularity 21 Nodes** ⁷

AMERICAN Covid 19

- **County Granularity 3126 Nodes**
 - **CVX** ⁵
 - **FISTA** ⁵
- **Province Granularity 51 Nodes** ⁵

PERSON TO PERSON Simulation (Population Size 500)

- **Random Selection Technique + Viral Load + No Recovery** ¹
- **Diverse Techniques + Viral Load + No Recovery** ⁵
- **Diverse Techniques + No Viral Load + Recovery (relaxed)** ⁵
- **Diverse Techniques + No Viral Load + Recovery (accelerated)** ⁵
- **Diverse Techniques with Log Weighted Lasso** ⁵

HPV Sim

- **Simulated and Reproduced the Plots from the Paper**

Figure 5.1: Tracks

5.0.2 Various Strategies for Subset Selection

Node Degree

The Idea here is to select that subset of Nodes that has largest possible degree. The hunch behind such a strategy is basically to filter the most influential nodes in the population set. Here the redish nodes represent the nodes with high node degrees

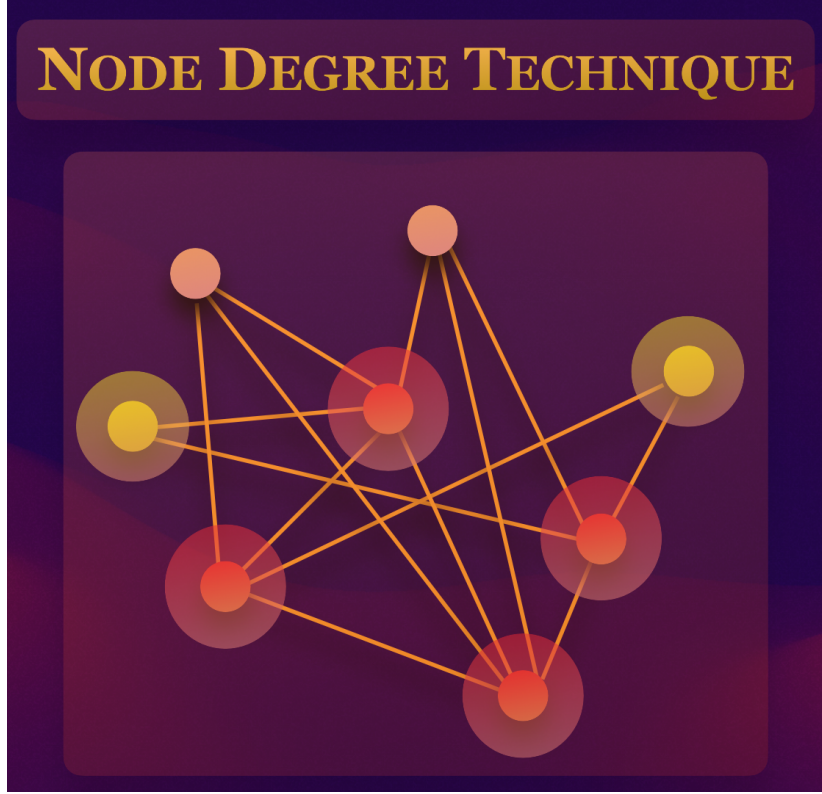


Figure 5.2: Node Degree

and yellow nodes represents nodes having degrees little lower.

Kemeny

This Technique is an inherently Greedy way of picking out subset of nodes. The Idea here is to get every node a score which will indicate how impact full its presence is in the graph. So here for doing that we will actually remove that node and then compute how the ‘movability’ is present in the Graph Now. Figure 5.3

Mutual Coherence

This technique also follows similar Greedish philosophy where in a ranking is given to all the nodes based on the Mutual Coherence scored calculated using the Sensing matrix and the Eigen Vector of the Laplacian, governed by this Rule: Minimize $\mu_i = \max_{i,j} |(\phi \cdot V)_{i,j}|$. There is also a notion of Average mutual coherence, which nothing but takes the average instead of max as in above. Results show that rather than Average, max works out a much better estimator to score nodes.

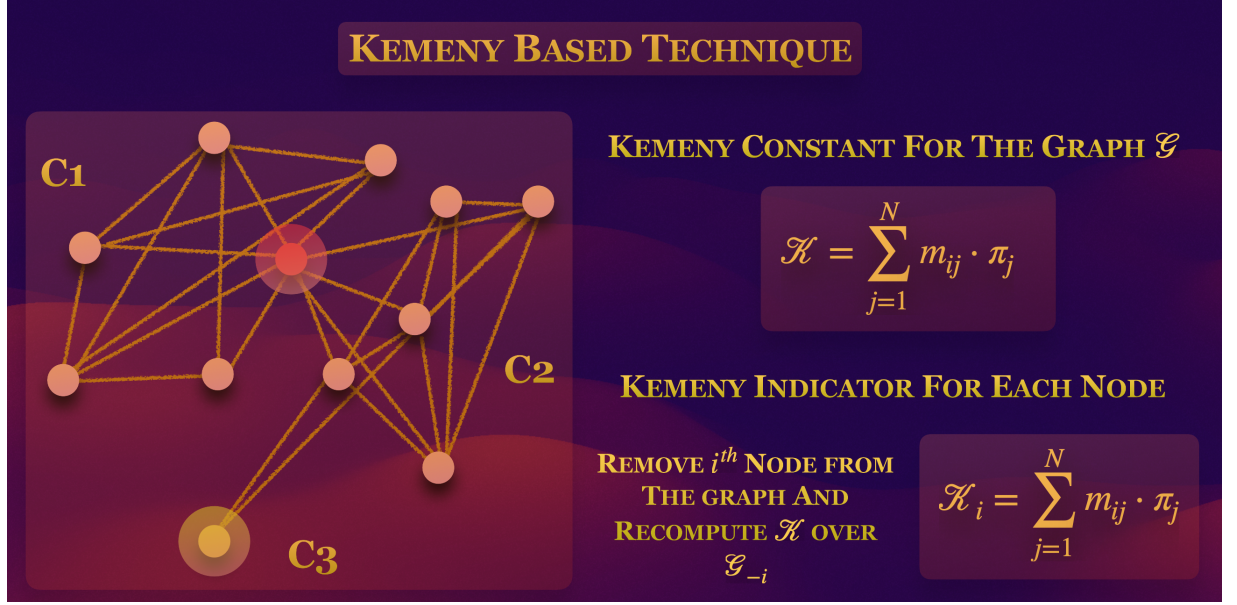


Figure 5.3: Kemeny Technique

Combination of Various techniques

In this regime, we go onto combine various subset selection strategies i.e. to pick some fraction of nodes by strategy A and the other fraction by strategy B, etc. This type of subset selection technique shows performance improvement.

5.1 Setting up of Model

Here We present the various settings that needs to be done to apply our model including setup for Cross Validation for picking up of data-driven Hyperparam selection. Below we show the protocol followed to perform cross-validation. Figure 5.4 Here We also perform an Adjusted Cross Validation Subset creation for Reconstruction, where I mix up and permute 80% of Non Zeros with Zero Health Status of various entities in the population set. The above mixing pattern ensures that the Cross-Validation based selection of hyperparam is being performed over a pragmatic setting which could possibly be in a realistic setting. Figure 5.5, Figure 5.6

5.1.1 BMC Measles Dataset

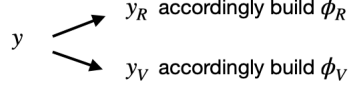
Here we analyzed in various settings, how could we do efficient subset selection. So we analysed this dataset in various dimensions, on the basis of Node-level Granularity where a node could be a Village or a hospital or a person.

1. Node Level analysis

Details regarding setting up of the model including cross-validation setting for picking up the hyper-parameters, i have provided above. Now in this section based on the fixed hyper-params i will present results with various techniques and their combination. Figure 5.7, Figure 5.8

Usage of Cross Validation to estimate the value for λ

Before Beginning with the Cross validation we build the y_R , ϕ_R , y_V , and ϕ_V from the y and the ϕ built initially.



For λ range starting from $1e-9$, $1e-8$.. till $1e+9$:

- Solve the Convex optimisation Problem using measurements from the Reconstruction R set.

$$\min_{\theta} \|y_R - \phi_R \cdot \mathcal{V} \cdot \theta\|_2^2 + \lambda_i \|\theta\|_1$$

$$\text{s.t. } 0 \leq V\theta \leq 1$$

- From above we obtain θ_{λ_i} , which we use to compute the Validation Errors for each λ_i .

$$VE_i = \min_{\theta} \|y_V - \phi_V \cdot \mathcal{V} \cdot \theta\|_2^2$$

Figure 5.4: Cross-Validation Protocol

2. Hospital-Level Analysis

Based on the cross-validation hyperparams are selected and then we apply our model over ONLY Hopsital and the corresponding results are shown below: Figure 5.9

3. Village-Level Analysis

Based on the cross-validation hyper params are selected and then we apply our model over ONLY Village and the corresponding results are shown below: Figure 5.10

5.1.2 BMC Measles Data From WHO

So here we first consider the scenario where we will have an analysis of nodes, which include, the Patients and the Hospitals and the Villages as separate entities and do modeling on them, together considered in a single Graph. Figure 5.11

1. Node Level Granularity

Now we will analyse the model performance over the Whole population containing Patients and Hospitals and villages. Figure 5.12

2. Ward Level Granularity

Here we consider at a zoomed-out level, a look at the whole population set divided in Communities as Ward. Figure 5.13

Kemeny + MC + Tuning Lambda with Adjusted Cross Validation Subset								
Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	10	6210.7335	0.3515	1	6406.0586	0.2551	195.33	0.10
300	10	6227.8715	0.3317	1	6408.6848	0.2306	180.81	0.10
350	10	6213.6477	0.3223	1	6422.9220	0.2106	209.27	0.11
370	10	6274.5631	0.3183	1	6333.8411	0.2026	59.28	0.12
400	1	6471.0417	0.1775	0.1	6603.6012	0.1745	132.56	0.00
450	1	6375.1552	0.1583	1E-09	6592.0061	0.1510	216.85	0.01
500	1E-08	6332.5379	0.1000	1E-09	6338.8620	0.0999	6.32	0.00
AVERAGE			0.2514			0.1892	142.92	0.06
Lambda Chosen which Minimises Cross Validation Error			Lambda Chosen which Minimises RMSE			Difference due to CHOICE of Lambda		

Node Degree + MC + Tuning Lambda with Adjusted Cross Validation Subset								
Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	1E-07	7436.8451	0.2599	1	7527.4206	0.2553	90.58	0.00
300	1E-09	7394.9857	0.2348	1	7504.9050	0.2297	109.92	0.01
350	10	7557.9258	0.3227	1	7634.3165	0.2099	76.39	0.11
370	1E-09	7330.3096	0.2047	1	7408.8402	0.2033	78.53	0.00
400	1E-09	7275.3508	0.1737	0.01	7327.7095	0.1735	52.36	0.00
450	1	7423.3918	0.1588	1E-09	7461.1783	0.1521	37.79	0.01
500	1E-09	7286.5961	0.0998	1E-09	7286.5961	0.0998	0.00	0.00
AVERAGE			0.2078			0.1891	63.65	0.02
Lambda Chosen which Minimises Cross Validation Error			Lambda Chosen which Minimises RMSE			Difference due to CHOICE of Lambda		

MC + Tuning Lambda with Adjusted Cross Validation Subset								
Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	10	3007.4391	0.3494	1	3037.8568	0.2559	30.42	0.09
300	10	3022.3485	0.3316	1	3068.4093	0.2283	46.06	0.10
350	10	3041.0553	0.3219	1	3145.0702	0.2098	104.01	0.11
370	10	3113.4056	0.3184	0.1	3374.6716	0.1968	261.27	0.12
400	10	3438.7900	0.3153	0.1	3655.5488	0.1726	216.76	0.14
450	10	3436.4082	0.3104	1E-09	3632.9500	0.1510	196.54	0.16
500	10	3500.3987	0.3031	1E-09	3706.7796	0.0999	206.38	0.20
AVERAGE			0.3214			0.1877	151.63	0.13
Lambda Chosen which Minimises Cross Validation Error			Lambda Chosen which Minimises RMSE			Difference due to CHOICE of Lambda		

Figure 5.5: Cross-Validation for Hyperparam Selection in Various Strategies for Subset Selection

5.1.3 American Covid-19 Dataset

This analysis we are doing now is based on the dataset we procured from John's Hopkins University, regarding the Covid-19 spread in the USA. Here we also take in to account the mobility data from US Govt Official Website where in a detailed cross-county and intra county mobility data records is given.

1. County Granularity

So we divide whole USA in to various counties, and build a fresh contact tracing map among the counties, which now takes in to account the travel from one place to another. We formulate the edge weight based on Gaussian RBF formula, quite common in Literature and show case also the limitations of usage of this generalised edge weight. Figure 5.14

2. Province Granularity

We now take in a more zoomed out, outlook on the US counties by grouping them within the provinces and thus analyse how the reconstruction is obtained here. Figure 5.15

Random					MC				
Measurements	MSE	Min Difference	Max Difference	Avg. Difference	Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	56.0792	7.269E-11	154.7108	1.1086	250	4.3173	5.748E-10	16.0000	1.0851
300	7.0395	5.794E-11	38.9384	0.8353	300	3.4380	4.160E-10	16.0000	0.8466
350	52.1276	2.354E-11	154.2275	0.8365	350	3.0123	1.088E-10	15.9890	0.5530
370	4.1162	3.357E-09	18.5366	0.8109	370	2.5733	4.470E-11	15.9340	0.4778
400	45.5331	5.119E-11	154.4227	0.5562	400	1.9685	1.958E-11	15.8753	0.3929
450	4.5915	1.150E-10	38.9836	0.3266	450	1.5092	7.733E-09	16.0000	0.2423
500	1.1983	1.481E-10	20.7327	0.1271	500	0.6598	8.802E-09	13.0000	0.0993

Node Degree					Node Degree + MC				
Measurements	MSE	Min Difference	Max Difference	Avg. Difference	Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	13.7607	1.536E-10	43.9872	1.4186	250	4.4734	7.420E-11	15.9927	0.9654
300	10.5033	2.725E-10	43.9805	0.9841	300	3.4785	6.051E-10	16.0000	0.8646
350	9.9197	2.400E-09	44.0000	0.8449	350	3.0063	1.314E-10	15.9127	0.5614
370	9.5501	8.988E-09	44.0000	0.7339	370	2.7629	4.019E-11	15.9416	0.5043
400	9.2461	3.810E-11	43.9908	0.6113	400	1.9935	2.570E-09	15.9958	0.3979
450	8.7739	1.561E-09	44.0000	0.7575	450	1.5325	1.582E-08	16.0000	0.2537
500	7.6384	2.460E-12	43.4076	0.3191	500	0.6575	1.176E-09	13.0000	0.0969

Kemeny					Kemeny + MC				
Measurements	MSE	Min Difference	Max Difference	Avg. Difference	Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	46.6021	6.780E-11	154.2739	0.9199	250	4.4407	7.121E-11	15.8394	0.9356
300	45.7964	4.694E-12	154.0280	0.7425	300	3.7075	2.030E-12	15.9926	0.6970
350	45.2192	5.520E-11	154.1698	0.6026	350	3.0742	9.500E-12	15.8912	0.5666
370	1.0389	2.961E-09	8.0611	0.3615	370	2.8160	2.410E-08	16.0000	0.5077
400	0.8639	1.292E-10	8.0480	0.2873	400	2.0145	1.185E-10	15.8651	0.4095
450	0.5117	2.170E-08	5.4665	0.1853	450	1.5081	1.343E-08	16.0000	0.2422
500	0.1529	7.730E-12	5.4924	0.0605	500	0.6581	1.404E-10	12.9934	0.0993

Figure 5.6: Mean Squared Error for the Obtained Reconstruction via various strategies.

5.1.4 Analysis on Synthetically Generated Dataset

We do a probabilistic simulation of over synthetic data on a person-to-person level granularity to assess the performance of our model now. But from the simulation results we observe that the connectivity between the entities is not conducive for giving a good reconstruction. Thus resting to the common setting of Gaussian RBF edge weighting function may not be able to capture the significance of connected ness. Figure 5.16

Log Weighted lasso Formulation

Measurements	Random	Node Degree	Kemeny	MC	AC	Node Degree + MC	Kemeny + MC	Node Degree + AC	Kemeny + AC	True Greedy MC
250	0.9222	0.4568	0.8407	0.2559	0.9037	0.2605	0.2595	0.9048	0.9078	0.9154
300	0.3267	0.3991	0.8334	0.2283	0.9071	0.2297	0.2371	0.9068	0.9046	0.2840
350	0.8891	0.3878	0.8281	0.2137	0.9054	0.2135	0.2159	0.9050	0.9011	0.2791
370	0.2498	0.3806	0.1255	0.1975	0.3865	0.2047	0.2066	0.9034	0.9007	0.2691
400	0.8310	0.3744	0.1145	0.1728	0.3625	0.1739	0.1748	0.3625	0.3618	0.2231
450	0.2639	0.3648	0.0881	0.1513	0.3483	0.1524	0.1512	0.3482	0.3497	0.2221
500	0.1348	0.3403	0.0481	0.1000	0.3222	0.0999	0.0999	0.3222	0.3222	0.1751

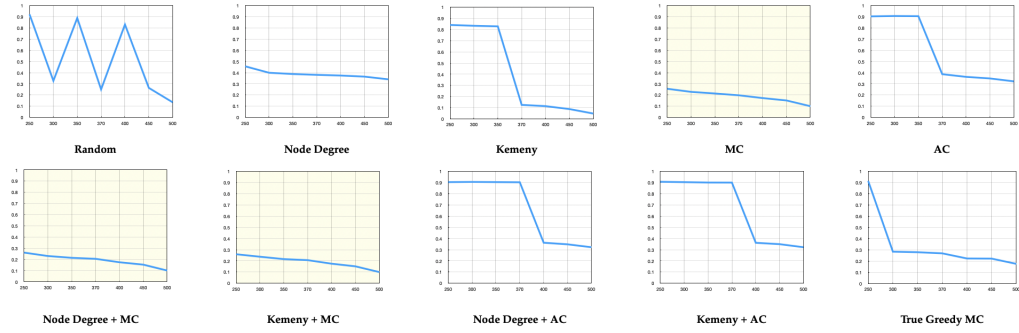


Figure 5.7: Results for Various Techniques.

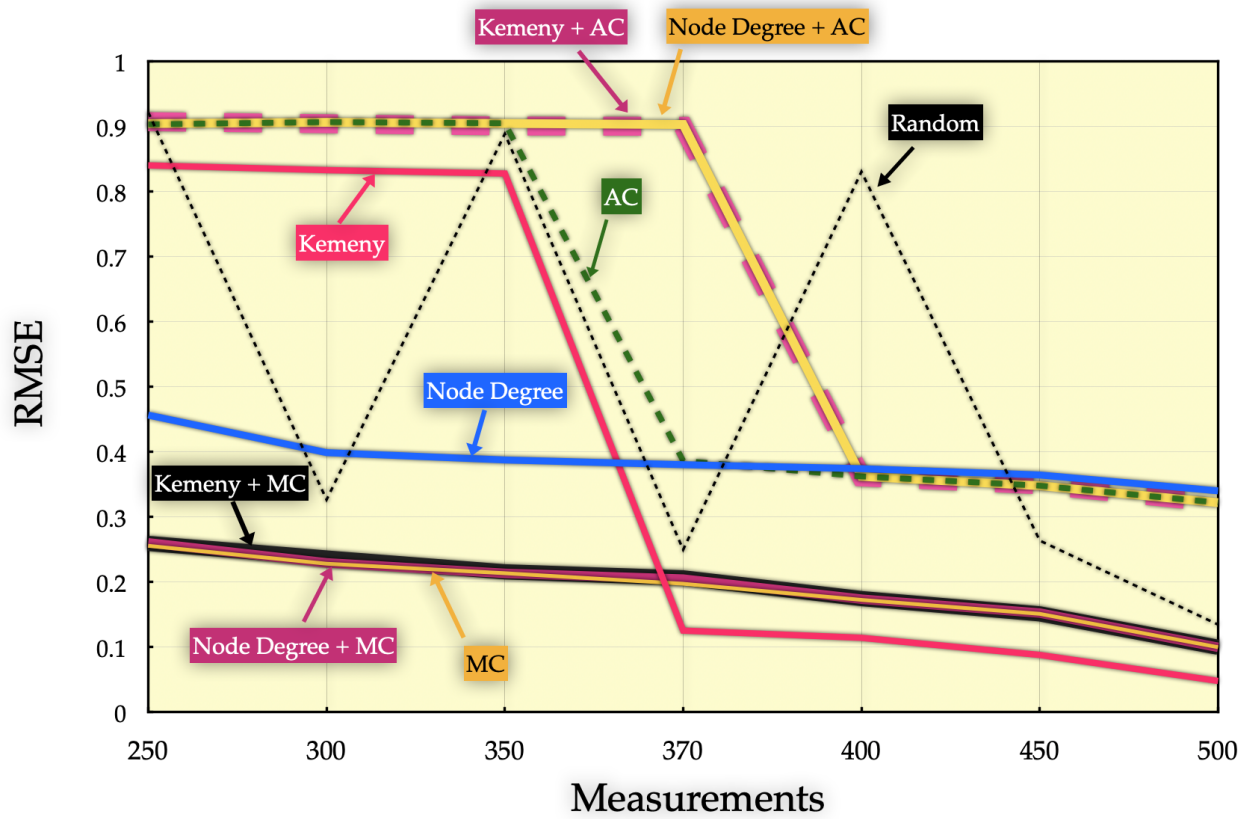


Figure 5.8: Comparative Plot for All Techniques in Reducing RMSE

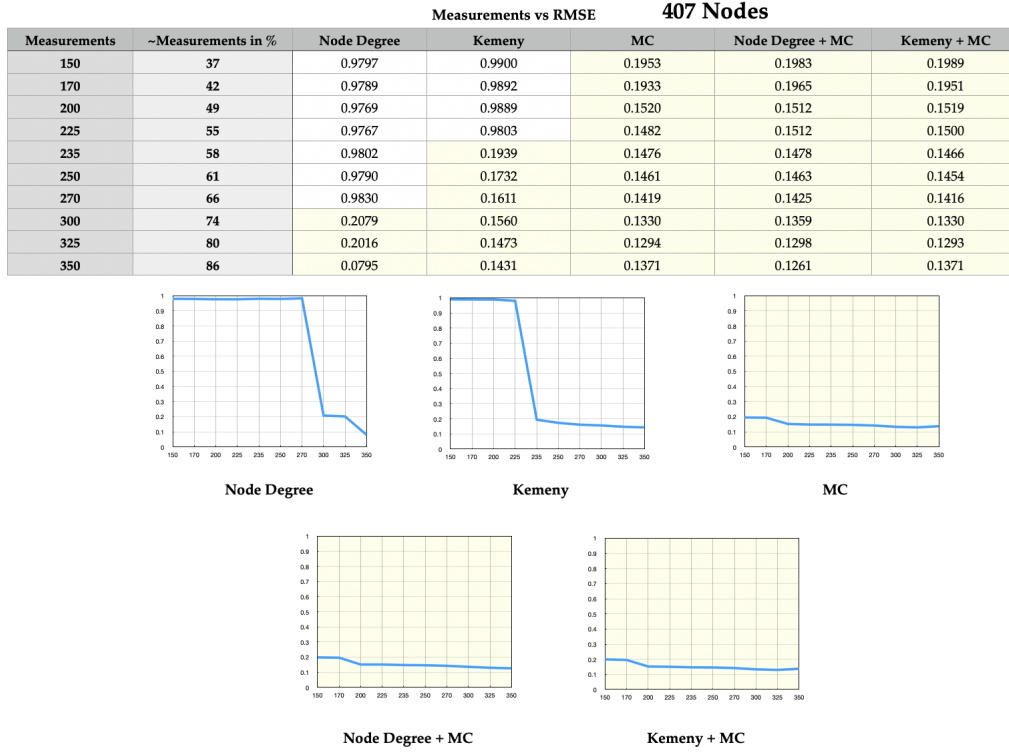


Figure 5.9: Hospital Level Nodes only Taken into consideration

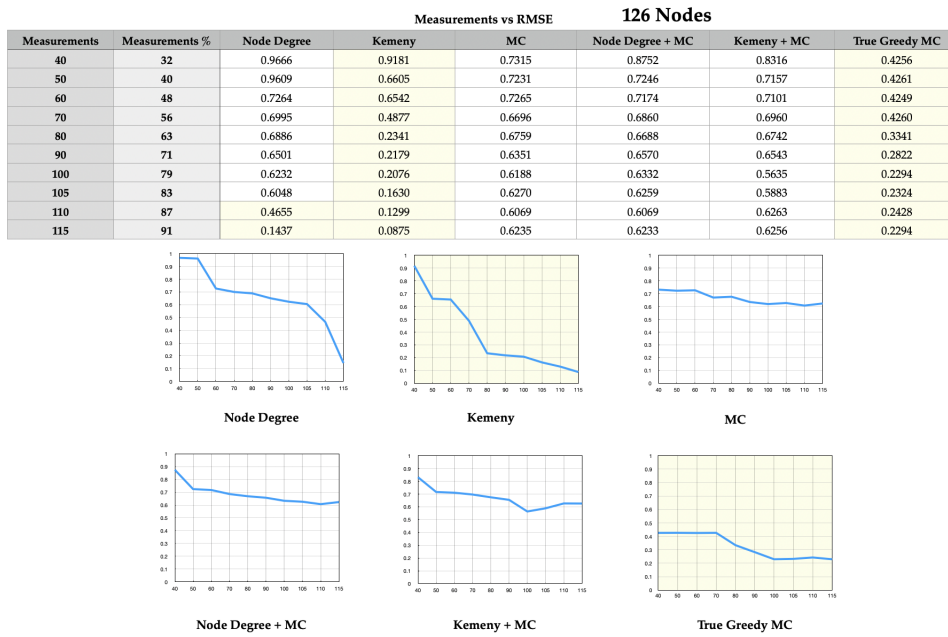


Figure 5.10: Village Level Nodes only Taken into consideration

BMC LATEST
Measurements vs RMSE

Measurements	Random	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
125	0.7766	0.4869	0.7874	0.3835	0.3870	0.3791	0.6896
150	0.3458	0.4420	0.2296	0.3655	0.3719	0.3207	0.6732
175	0.7155	0.3803	0.1921	0.3616	0.3617	0.3463	0.6416
200	0.2736	0.3389	0.1921	0.3593	0.3599	0.3463	0.6013
225	0.2726	0.2742	0.1643	0.2687	0.2689	0.2687	0.5692
250	0.1726	0.2111	0.1278	0.1157	0.1175	0.1060	0.5355
270	0.1123	0.1229	0.1264	0.0918	0.0918	0.0918	0.5346
275	0.0849	0.1183	0.1072	0.0876	0.0916	0.0876	0.5337
300	0.0984	0.0741	0.0899	0.0655	0.0655	0.0655	0.5283

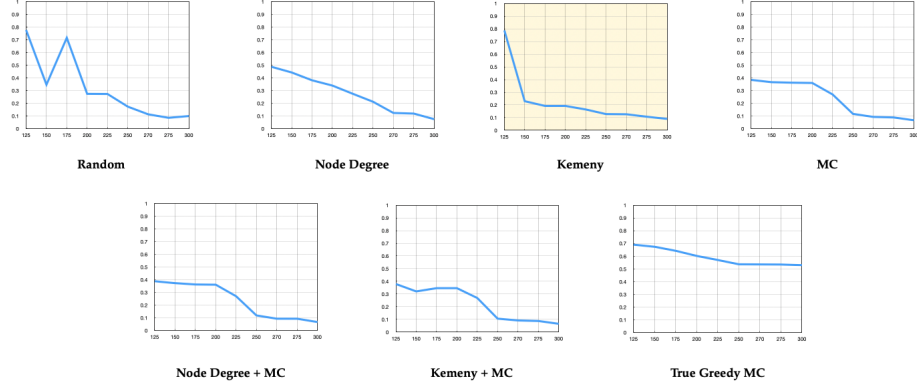


Figure 5.11: Node Level Granularity

Measurements vs MSE

Measurements	Random	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
125	3330.92	1309.60	3424.24	812.14	827.18	793.61	2626.57
150	660.26	1078.76	2994.78	737.87	763.99	567.97	2503.15
175	2827.24	798.96	291.04	722.03	722.38	662.24	2273.32
200	413.51	634.15	203.89	713.09	715.24	662.28	1996.65
225	410.38	415.36	149.08	398.79	399.34	398.81	1789.32
250	164.49	246.23	90.23	73.93	76.26	62.08	1583.54
270	69.70	83.43	88.20	46.52	46.53	46.52	1578.44
275	39.76	77.28	63.51	42.40	46.38	42.40	1573.34
300	53.52	30.33	44.61	23.67	23.67	23.67	1541.75

Figure 5.12: Measurement Vs MSE reconstruction Analysis

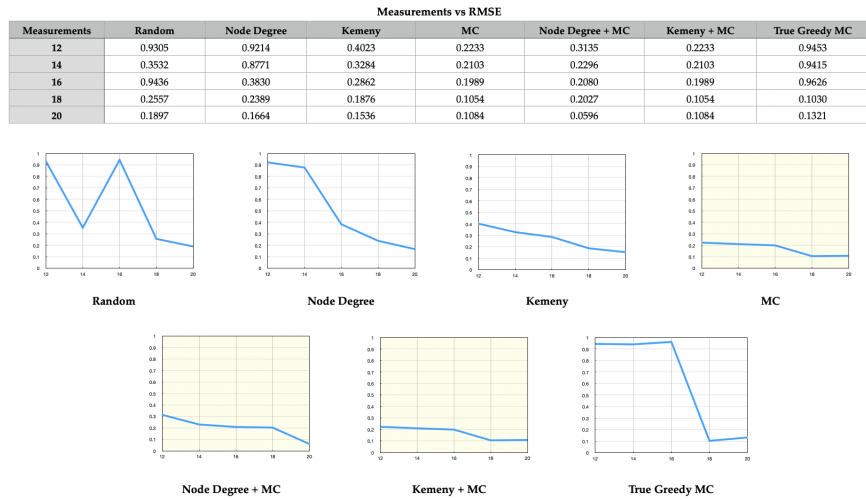
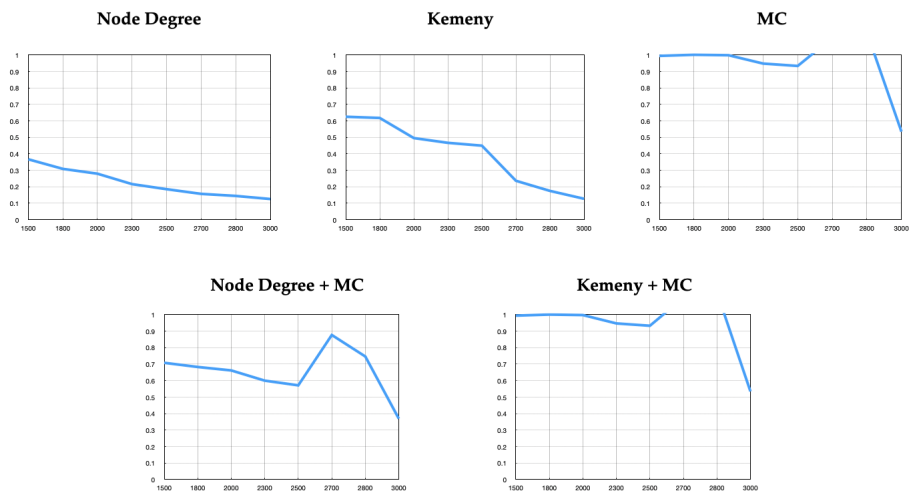


Figure 5.13: Ward Level Nodes only Taken into consideration

Measurements vs RMSE					
Measurements	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC
1500	0.3662	0.6245	0.9938	0.7078	0.9943
1800	0.3080	0.6171	1.0007	0.6825	1.0007
2000	0.2788	0.4950	0.9979	0.6616	0.9976
2300	0.2156	0.4662	0.9472	0.5996	0.9469
2500	0.1848	0.4490	0.9328	0.5710	0.9328
2700	0.1563	0.2358	1.1031	0.8767	1.1031
2800	0.1439	0.1744	1.1361	0.7462	1.1361
3000	0.1251	0.1266	0.5341	0.3686	0.5341



**US DATASET
COUNTY GRANULARITY
FISTA BASED**

Figure 5.14: US Counties Level granularity

Measurements vs RMSE					
Measurements	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC
20	0.5278	0.8484	0.7125	0.7958	0.6649
25	0.6142	0.7928	0.7084	0.6769	0.6476
30	0.4980	0.6155	0.3963	0.7018	0.4712
35	0.2211	0.5673	0.3886	0.3873	0.3818
40	0.2781	0.2222	0.2269	0.2171	0.3724
45	0.1176	0.1734	0.1804	0.1727	0.2228

US DATASET

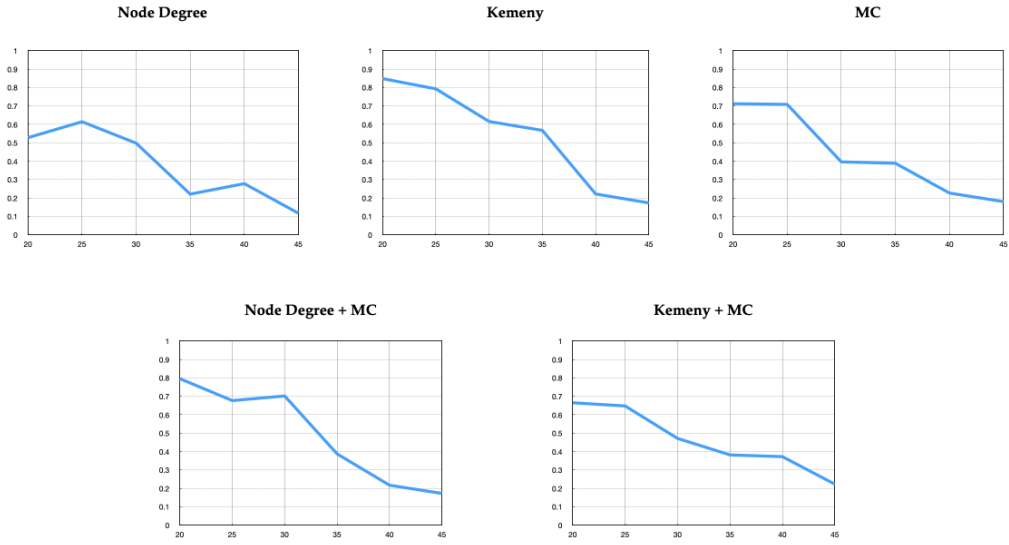


Figure 5.15: US Province Level granularity

$$\begin{aligned}
 & \min_{\mathbf{x}: \mathbf{y} = \mathbf{A}\mathbf{x}} \sum_{i=1}^N (-\log p_i) |x_i| \\
 & \min_{\mathbf{x}} \left(\|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2 + \tau \sum_{i=1}^N (-\log p_i) |x_i| \right)
 \end{aligned}$$

Figure 5.16: Formulation used for Person to Person Simulation

Graph Learning

Section 1 Various Future Advancements

Two tracks for future advancements:

1. **Learning**/Inferencing the **Graph Laplacian** from the data.
2. Analysing and Contrasting between **Retrospective** and **Prospective** perspectives in vaccination.

Details of each written above:

1. Learning/Inferencing the Graph Laplacian from the data
 - Discussed a wide array of methods from statistical, physical, and GSP perspectives and then looked at the various cost functions.
 - One important exploration was relating the norms, that were imposed in various cost functions:
 - L1 Norm Penalty:** Ensures global sparsity, yet may lack row-wise sparsity. *False sense of Sparsity.*
 - L2 Norm Penalty:** Risks voiding rows, Promoting disconnection in the graphs and *thus loss of contact Information nullifying the whole graph learning prospect.*
 - Need to craft a fresh approach to balance trade-offs, achieving desired graph sparsity for improved reconstruction
2. Analysing And contrasting between Retrospective and Prospective perspectives in vaccination.

Retrospective Perspectives

- Retrospective perspectives on vaccination involve looking back at the history and impact of vaccination efforts. This includes examining the development of vaccines, their effectiveness in preventing diseases, their role in public health, and any lessons learned from past vaccination campaigns. Understanding the successes and challenges of vaccination in hindsight can help inform future strategies and improve global health outcomes.
- This could involve looking back at the various contacts a person have had and then taking vaccination decision against them.

Prospective Perspectives

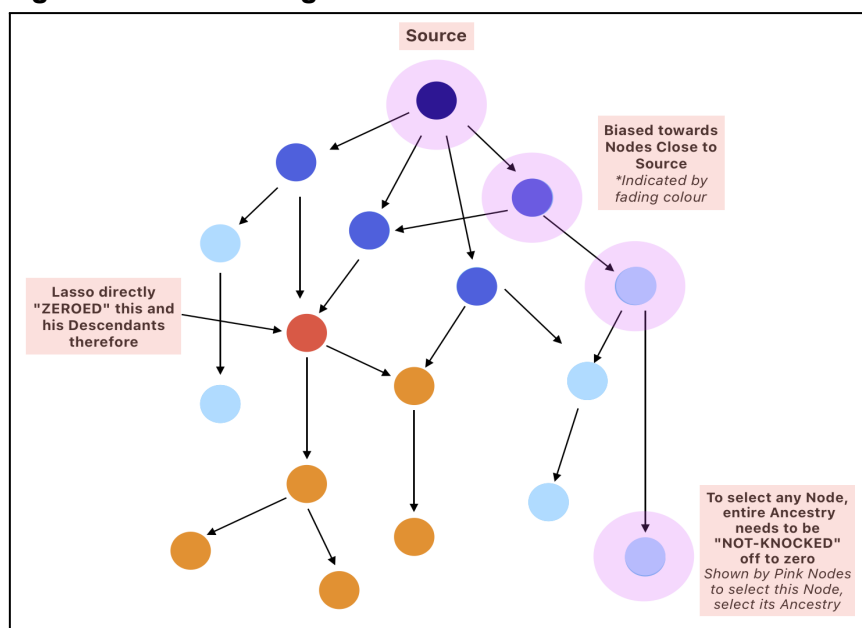
- A prospective perspective in vaccination refers to looking forward and anticipating the potential impact of upcoming vaccination efforts. This involves considering factors such as the development of new vaccines, their distribution, potential challenges in reaching target populations, and the expected benefits for public health. Examining vaccination from a prospective standpoint helps in planning and strategizing for effective and efficient vaccination campaigns to prevent and control diseases.
- This could involve looking at the neighbors of the person. Now the person may not necessarily have had contact with those neighbours. Then make a decision about whom to vaccinate.

Section 2 Analysis of Various Formulations and L1/Rho Block Norm Regulariser.

- The theme of the discussion briefly centered on strategizing approaches to effectively address the **tradeoff between L1 and L2 norms** to rightly **capture the sparsity**.
- Analysis of the Convex Optimization formulation applied in Statistical techniques, particularly the **Graphical LASSO**, unveiled that the **L1 norm imposed** over the Inverse Covariance Matrix **doesn't appear** to impose **sparsity row-wise**, which is pivotal. The formulation is shown in the image below:

$$\begin{array}{ll} \underset{\Theta, \sigma^2}{\text{maximize}} & \log \det \Theta - \text{tr} \left(\frac{1}{M} \mathbf{X} \mathbf{X}^T \Theta \right) - \rho \|\Theta\|_1, \\ \text{subject to} & \Theta = \mathbf{L} + \frac{1}{\sigma^2} \mathbf{I}, \mathbf{L} \in \mathcal{L}, \end{array}$$

- Discussed the idea of imposing an **L1/L_q Block Norm Regulariser (gHKL)** $q \in (1, 2)$ instead, which is presented as an **upgrade** over conventional **L1/L2 regulariser (HKL)** in the Ramakrishnan et al. “**Generalized Hierarchical Kernel Learning**” Paper.
- The problem discussed in the paper is about **learning a composite kernel** as a **sparse linear combination** of many base **kernels**, for the classification task where Base Kernels are represented in the form of a **DAG** as in the traditional Bach. (2008) implementation, thus carrying forward this set-up, the **gHKL** framework mentioned in the paper mitigates the challenges that come inherently with the posed HKL framework, by imposing a more “**liberal**” regulariser i.e. **L1/L_q $q \in (1, 2)$ Block Norm Regulariser**.
- Key Challenges with Bach(2008). HKL:
 - a) A **kernel** is selected **only** if the kernels associated with all its **ancestors** in the **DAG** are **selected**.
 - b) The **weight** of the **kernel** associated with a (selected) **node** will **always** be **greater** than the **weight** of the **kernels** associated with its **descendants**.



- Letting ρ go free hand between 1 and 2 imparted flexibility and thus helped mitigate the above challenges as in the traditional HKL.
- The formulation of gHKL is given below:

$$\Omega_S(f) = \sum_{v \in \mathcal{V}} d_v \|f_{D(v)}\|_\rho,$$

$$\text{where } f = (f_v)_{v \in \mathcal{V}}, \|f_{D(v)}\|_\rho = \left(\sum_{w \in D(v)} \|f_w\|^\rho \right)^{\frac{1}{\rho}} \text{ and } \rho \in (1, 2]$$

Comment on the above:

- Since the l1-norm promotes sparsity, it follows that $\|f_{D(v)}\|_\rho = 0$ (that is $f_w = 0 \forall w \in D(v)$) for most $v \in \mathcal{V}$. This phenomenon is similar to HKL. But now, even in cases where $\|f_{D(v)}\|_\rho$ is not forced to zero by the l1-norm, many components of $f_{D(v)}$ tend to zero (that is $f_w \rightarrow 0$ for many $w \in D(v)$) as the value of ρ tends to unity. Also, note that $\rho = 2$ renders the HKL regularizer.
- To summarize, the proposed gHKL formulation is

$$\min_{f_v \in \mathcal{H}_{k_v}, \forall v \in \mathcal{V}, b \in \mathbb{R}} \frac{1}{2} (\Omega_S(f))^2 + C \sum_{i=1}^m \ell(y_i, F(\mathbf{x}_i)).$$

- Careful investigation of the approach by Dong et al. based on GSP Techniques, to learn a graph based on a smoothness assumption of signals leads to the conclusion that the “Elastic-Net” like Regularization chosen there doesn't impose row-wise sparsity but rather imposes a group sparsity in the sense that the highly correlated variables get similar regression coefficients, which is NOT what is desired.

Conclusion of the Above Discussion:

- Discussed mathematical details of Graphical Lasso formulation (Log-Likelihoods and the Penalty terms) and various constraints imposed on the search space of inverse covariance matrix, pivotal for learning the factorization of the joint probability distribution over the variables as well as the underlying graphical model structure (edges and their associated weights). Also Discussed the rest of the statistical Techniques.
- Discussed the limitations of the Statistical techniques.
- Then moved to the GSP Techniques, where discussed extensively learning graphs over which the signals would have a “smoothness” notion, defined by the Laplacian Quadratic term. (and how it was also implicitly getting captured in the statistical techniques as well)
- Discussed Dong et al's formulation. Also pointed out the group sparsity is not getting captured in the desired sense (sparsity) in this formulation, but rather captured in some other sense. (correlated variable to have similar regression coefficients)

$$\begin{aligned}
& \min_{L \in \mathbb{R}^{n \times n}, Y \in \mathbb{R}^{n \times p}} \|X - Y\|_F^2 + \alpha \text{tr}(Y^T L Y) + \beta \|L\|_F^2 \\
& \text{s.t. } \text{tr}(L) = n, \\
& \quad L_{ij} = L_{ji} \leq 0, i \neq j, \\
& \quad L \cdot \mathbf{1} = \mathbf{0},
\end{aligned}$$

- Discussed in detail the formulation of the gHKL framework as an upgrade over HKL and MKL and how L1/Lp formulation gives “liberal” regularisation to capture sparsity in a block-wise fashion.
- Analysis of **Kalofolias** Formulation:

Here the goal is to give a general-purpose model for learning graphs when no prior information is available. In order to obtain meaningful graphs, they want to make sure that each node has at least one edge with another node.

Further, they add, It is also desirable to have control of how sparse is the resulting graph. To meet these expectations, they propose the following model with parameters $\alpha > 0$ and $\beta \geq 0$ controlling the shape of the edges:

$$\underset{W \in \mathcal{W}_m}{\text{minimize}} \quad \|W \circ Z\|_{1,1} - \alpha \mathbf{1}^T \log(W \mathbf{1}) + \beta \|W\|_F^2$$

Adding a logarithmic term ($\beta = 0$) solely leads to very sparse graphs, and changing **alpha** only changes the scale of the solution and not the sparsity pattern. For this reason, they add the third term.

Adding an L-1 norm to control sparsity is not very useful. On the other hand, by adding a Frobenius norm we penalize the formation of big edges but do not penalize smaller ones. This leads to more dense edge patterns for bigger values of β .

Section 3 Our Proposed Formulation

The formulation exhibits certain characteristics and potential issues that should be addressed:

$$\begin{aligned}
& \text{Minimize} \quad J(V, \Delta, \alpha) = \sum_{t=1}^T \|\tilde{\mathbf{x}}(t) - V \tilde{\boldsymbol{\alpha}}(t)\|_2^2 + \mu_1 \cdot \|\tilde{\boldsymbol{\alpha}}(t)\|_1 - \mu_2 \cdot \mathbf{1}^T \log(V \Delta V^T \mathbf{1}) + \mu_3 \cdot \|V \Delta V^T\|_1 \\
& \text{subject to} \quad V^H V = I^F \\
& \quad V[:, 1] = \mathbf{1} \\
& \quad \Delta[1, 1] = 0
\end{aligned}$$

- **L1 Norm vs. L1 Rho Norm:** The use of the L1 norm on variable W can lead to sparsity in the solution whereas the log term discourages disconnectivity by not letting the degrees of any node be zero. Thus with Rho as a hyperparameter, the L1 Rho norm can provide more flexibility and control over the sparsity level, potentially addressing the adversarial behavior issue compared to a vanilla L1 norm.

- **Mirror Descent:** To solve the optimization problem effectively with the L1 Rho norm and other constraints, it is mentioned that a Mirror Descent-like algorithm may be necessary. This is a valid approach, as Mirror Descent is often used for non-smooth convex optimization problems with constraints.

3.3 gHKL_{MT} Dual Formulation

As mentioned earlier, due to the presence of the ℓ_ρ -norm term in gHKL_{MT} formulation, naive extensions of the projected gradient based active set method in Bach (2008) will be rendered computationally infeasible on real world data sets. Hence, we first re-write gHKL_{MT} formulation in an elegant form, which can then be solved efficiently. To this end, we note the following variational characterization of $\Omega_T(f_1, \dots, f_T)$.

- **Symmetry, Orthonormality, and Positive Semidefiniteness:** It's important to explicitly impose constraints on relevant matrices to ensure desirable properties. This includes enforcing symmetry, orthonormality, and positive semidefiniteness, depending on the specific matrices involved in the problem.
- **Edge Pruning:** Zeroing out edges that are below a certain threshold. This threshold should be considered a hyperparameter and can be part of the regularization strategy. Pruning small edges can lead to a more interpretable and efficient graph representation.
- **Signal Filtering:** Applying a filter to the input signals X is suggested. Filtering can help improve the convergence of the optimization problem and lead to the desired output. The choice of the filter and its parameters should be considered carefully based on the characteristics of the signals and the problem's objectives.

In summary, the formulation can be enhanced by considering an L1 Rho norm, employing Mirror Descent or a similar optimization algorithm, explicitly imposing relevant matrix constraints, tuning the edge pruning threshold as a hyperparameter, and applying signal filtering for improved convergence and results. These considerations can lead to a more robust and effective solution for the problem at hand.

Section 4 Update 1 to the Formulation: Caping Delta via ICM

Updated Formulation: Impose a Cap on Delta via the inverse of Covariance Matrix.

$$\begin{aligned}
 &\text{Minimize } J(V, \Delta, \alpha) = \sum_{t=1}^T \|\tilde{\mathbf{x}}(t) - V\tilde{\boldsymbol{\alpha}}(t)\|_2^2 + \mu_1 \cdot \|\tilde{\boldsymbol{\alpha}}(t)\|_1 - \mu_2 \cdot \mathbf{1}^T \log(V\Delta V^T \mathbf{1}) + \mu_3 \cdot \|V\Delta V^T\|_1 + \mu_4 \left\| \hat{\boldsymbol{\Theta}} - V\Delta V^T - \frac{1}{\sigma^2} I \right\|_2^2 \\
 &\text{subject to } V^H V = I^F \\
 &\quad V[:, 1] = \mathbf{1} \\
 &\quad \Delta[1, 1] = 0
 \end{aligned}$$

The above is solved by alternatingly minimizing in a triplet fashion as shown:

$$\begin{aligned}
 &\textbf{Step 1: } \min_{\alpha} \|X - V\alpha\|_F^2 + \mu_1 \|\alpha\|_1 \\
 &\text{subject to: } \alpha \in R^{(\text{numNodes} \times \text{numTimeSteps})} \\
 \\
 &\textbf{Step 2: } \min_{\Delta} \left(-\mu_2 \mathbf{1}^T \log(V\Delta V^T \mathbf{1}) + \mu_3 \|V\Delta V^T\|_1 + \mu_4 \left\| \hat{\boldsymbol{\Theta}} - V\Delta V^T - \frac{1}{\sigma^2} I \right\|_F^2 \right) \\
 &\text{subject to: } \Delta \in R^{(\text{numNodes}-1 \times \text{numNodes}-1)}, \text{ diagonal} \\
 \\
 &\textbf{Step 3: } \min_V \|X - V\alpha\|_F^2 - \mu_2 \mathbf{1}^T \log(V\Delta V^T \mathbf{1}) + \mu_3 \|V\Delta V^T\|_1 + \mu_4 \left\| \hat{\boldsymbol{\Theta}} - V\Delta V^T - \frac{1}{\sigma^2} I \right\|_F^2 \\
 &\text{subject to: } V \in R^{(\text{numNodes} \times \text{numNodes})}
 \end{aligned}$$

Section 5 Update 2 to the Formulation: Replacing term inside log with Degree sum.

Imposing the mew4 term, which was intended to cap the delta values, doesn't seem to do that, rather they promote the $V^* \Delta V^T$ term to get negative and thus the logarithmic term to become complex-valued.

So the decision is made to replace the $V^* \Delta V^T$ term with the abs sum of it.

Also about the signals regime, it is fixed to try out as an additional experiment, over 1 community and little stronger connections present within it, to contrast against the case of Stochastic Block Models SBMs, wherein we have multiple communities and sparse connections across communities.

The possible reason for this seems that the SBM with multiple communities might have multiple local minima as opposed to the one with a single community with a single minima.

Section 6 Update 3 to the Formulation. Augmenting with Manual gradients.

Updated Formulation: Here we are now placing log of diagonal values from $V \cdot \Delta \cdot V^\top$, within the cost function to prevent the disconnectivity and a hard constraint is being imposed on the rows of the $V \cdot \Delta \cdot V^\top$ to be equal to zero.

$$\begin{aligned} \text{Minimize } J(V, \Delta, \alpha) &= \sum_{t=1}^T \|\tilde{\mathbf{x}}(t) - V\tilde{\boldsymbol{\alpha}}(t)\|_2^2 + \mu_1 \cdot \|\tilde{\boldsymbol{\alpha}}(t)\|_1 - \mu_2 \cdot \mathbf{1}^\top \log(\text{diag}(V\Delta V^\top)) + \mu_3 \cdot \|V\Delta V^\top\|_1 + \mu_4 \left\| \hat{\boldsymbol{\Theta}} - V\Delta V^\top - \frac{1}{\sigma^2} I \right\|_2^2 \\ \text{subject to } V^H V &= I^F \\ V[:, 1] &= \mathbf{1} \\ \Delta[1, 1] &= 0 \\ V\Delta V^\top \mathbf{1} &= \vec{0} \end{aligned}$$

$$\text{Step 1: } \min_{\alpha} \|X - V\alpha\|_F^2 + \mu_1 \|\alpha\|_1$$

$$\text{subject to: } \alpha \in R^{(\text{numNodes} \times \text{numTimeSteps})}$$

$$\text{Step 2: } \min_{\Delta} \left(-\mu_2 \mathbf{1}^\top \log(\text{diag}(V\Delta V^\top)) + \mu_3 \|V\Delta V^\top\|_1 + \mu_4 \left\| \hat{\boldsymbol{\Theta}} - V\Delta V^\top - \frac{1}{\sigma^2} I \right\|_F^2 \right)$$

$$\text{subject to: } \Delta \in R^{(\text{numNodes}-1 \times \text{numNodes}-1)}, \text{ diagonal}, V\Delta V^\top \mathbf{1} = \vec{0}$$

$$\text{Step 3: } \min_V \|X - V\alpha\|_F^2 - \mu_2 \mathbf{1}^\top \log(\text{diag}(V\Delta V^\top)) + \mu_3 \|V\Delta V^\top\|_1 + \mu_4 \left\| \hat{\boldsymbol{\Theta}} - V\Delta V^\top - \frac{1}{\sigma^2} I \right\|_F^2$$

$$\text{subject to: } V \in R^{(\text{numNodes} \times \text{numNodes})}, V\Delta V^\top \mathbf{1} = \vec{0}$$

$$f = \|X - V \cdot A\|_2^2$$

$$\frac{\partial f}{\partial V} = -2 \cdot (X - V \cdot A) \cdot A^\top$$

$$f = -\mu_2 \cdot I^\top \cdot \log(\text{diag}(V \cdot \Delta \cdot V^\top))$$

$$\frac{\partial f}{\partial V} = -(\mu_2 \cdot \text{diag}(I \oslash \text{diag}(V \cdot \Delta \cdot V^\top)) \cdot V \cdot \Delta^\top + \mu_2 \cdot \text{diag}(I \oslash \text{diag}(V \cdot \Delta \cdot V^\top)) \cdot V \cdot \Delta)$$

$\oslash$: Element-wise-Divide

$$f = \mu_3 \cdot \|V \cdot \Delta \cdot V^\top\|_1$$

$$\frac{\partial f}{\partial V} = \mu_3 \cdot \text{sign}(V \cdot \Delta \cdot V^\top) \cdot V \cdot \Delta^\top + \mu_3 \cdot \text{sign}(V \cdot \Delta^\top \cdot V^\top) \cdot V \cdot \Delta$$

$$f = \mu_4 \cdot \left\| \hat{\boldsymbol{\Theta}} - V \cdot \Delta \cdot V^\top - \frac{1}{\sigma^2} \cdot I \right\|_2^2$$

$$\frac{\partial f}{\partial V} = -(2 \cdot \mu_4 \cdot (\hat{\boldsymbol{\Theta}} - V \cdot \Delta \cdot V^\top - \frac{1}{\sigma^2} \cdot I) \cdot V \cdot \Delta^\top + 2 \cdot \mu_4 \cdot (\hat{\boldsymbol{\Theta}}^\top - V \cdot \Delta^\top \cdot V^\top + (-\frac{1}{\sigma^2} \cdot I)^\top) \cdot V \cdot \Delta)$$

***Need to update this even More. Regarding the hard constraint imposed within Manopt code.

Code Flow:

```

[V, delta, U] = svd(X);
V(:, 1) = ones(size(V, 1), 1) / sqrt(size(V, 1));
[V, ~] = qr(V);
delta(1, 1) = 0;
theta_cap=inv((X*X')/(numTimeSteps-1));
[~, E]=eig(theta_cap);
sigma=min(diag(E));

```

```

cvx_begin quiet
    variable alpha(numNodes, numTimeSteps)
    minimize(sum(sum_square_abs(X - V * alpha)) + mew1 * norm(alpha(:), 1))
cvx_end

```

```

prevV=V;
V=V(:, 2:end);

```

```

cvx_begin quiet
    variable delta(numNodes-1, numNodes-1) diagonal
    minimize (-1*mew2*ones(1, numNodes)*log(diag(V*delta*V')) +
        mew3 * sum(sum(abs(V * delta * V')))) +
        mew4 * sum(sum_square_abs(theta_cap-V*delta*V'-(1/sigma^2)*eye(numNodes))))
    subject to
        V*delta*V'*ones(numNodes, 1) == zeros(numNodes, 1);
cvx_end

```

```

delta=diag([0; diag(delta)]);
V=prevV;

```

```

problem.M = stiefelfactory(numNodes, numNodes);
problem.cost = @(V) sum(sum_square_abs(X - V * alpha)) - mew2 * ones(1, numNodes) * log(diag(V*delta*V'))
    + mew3 * sum(sum(abs(V * delta * V')))) +
    + mew4 * sum(sum_square_abs(theta_cap-V*delta*V'-(1/sigma^2)*eye(numNodes)));
problem.nonlcon = @(V) norm(V*delta*V'*ones(numNodes,1));
problem.egrad = @(V) Compute_gradients(X,V,alpha,delta,mew2,mew3,mew4,theta_cap,numNodes,sigma);
[V, cost, info] = trustregions(problem);

```

```

V(:,1)=ones(numNodes,1)/sqrt(size(V,1));
[V, ~] = qr(V);

```

```

function grad=Compute_gradients(X,V,alpha,delta,mew2,mew3,mew4,theta_cap,numNodes,sigma)
    grad1 = -2*(X-V*alpha)*alpha';

    T0=V*delta;
    t1=ones(numNodes,1)./diag(T0*V');
    grad2 = mew2*diag(t1)*V*delta'+mew2*diag(t1)*T0;

    grad3=mew3*sign(V*delta*V')*V*delta'+mew3*sign(V*delta'*V')*V*delta;

    t0=2*mew4;
    T1=(1/sigma^2)*eye(numNodes);
    grad4=-(t0*(theta_cap-V*delta*V'-T1)*V*delta'+t0*(theta_cap'-V*delta'*V'+(-T1)')*V*delta);

    grad=grad1-grad2+grad3+grad4;
end

```

Section 7 Introspecting Hyper-params and Other Formulation.

1. Hyper-parameter Tuning:

μ_2 should be higher as it decides scale (doesn't impact SPARSITY PATTERN) and μ_3 is basically deciding the density of edges (impact SPARSITY PATTERN)

if i increase μ_3 then **More** will the **#of Edges** be in the **Learned Graph**

if i increase μ_2 then **More** will the **Scale of Edge Weight** be in the **Learned Graph**.

So these 2 are like TUNING KNOBS which are giving us the freedom to choose the AMOUNT of SPARSITY in the GRAPH

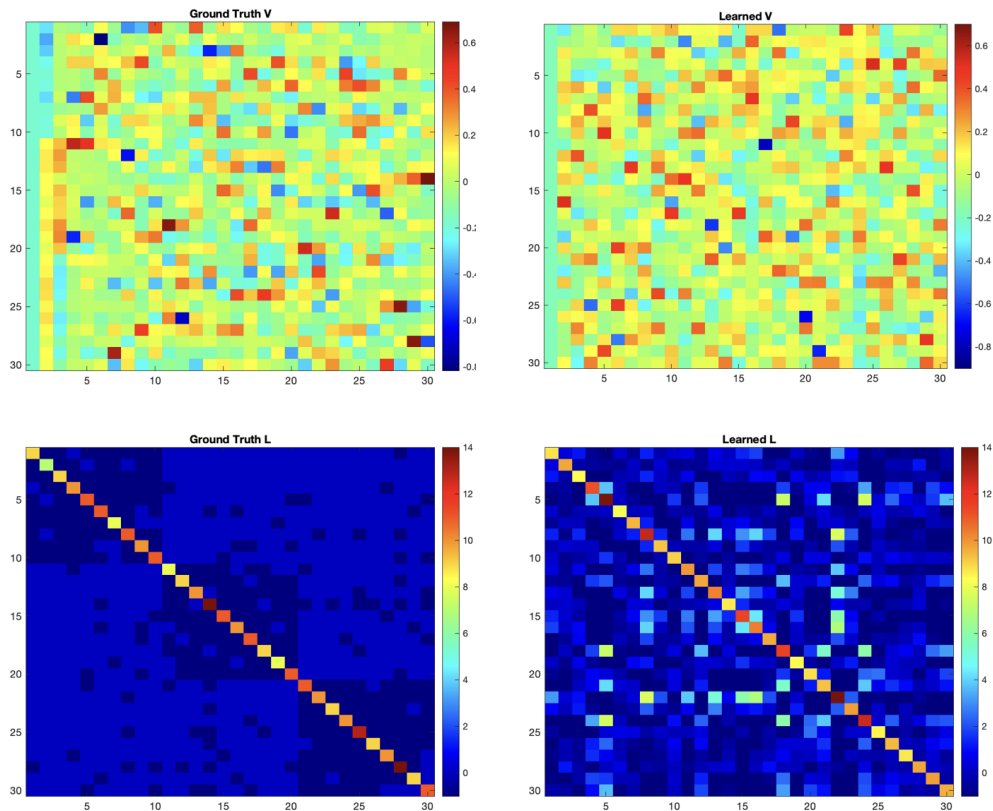
Grid Search analysis by Dong to quantify a relationship between mew2 and mew3.

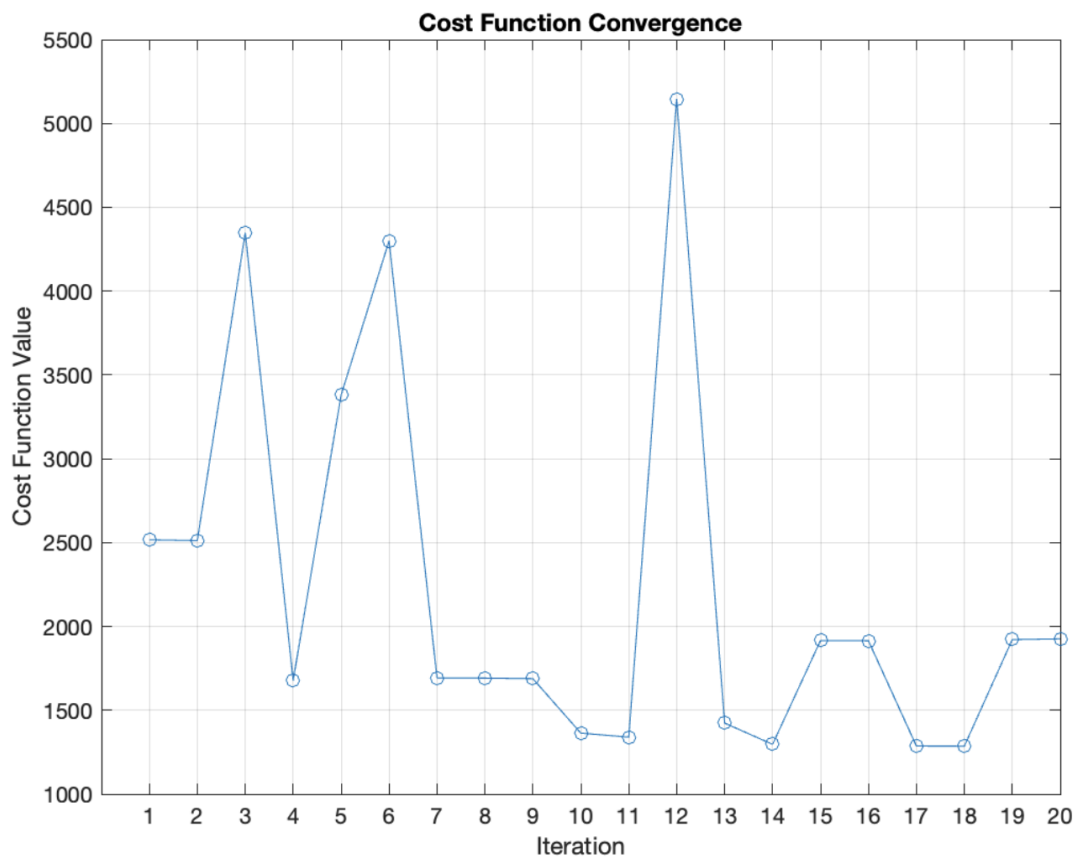
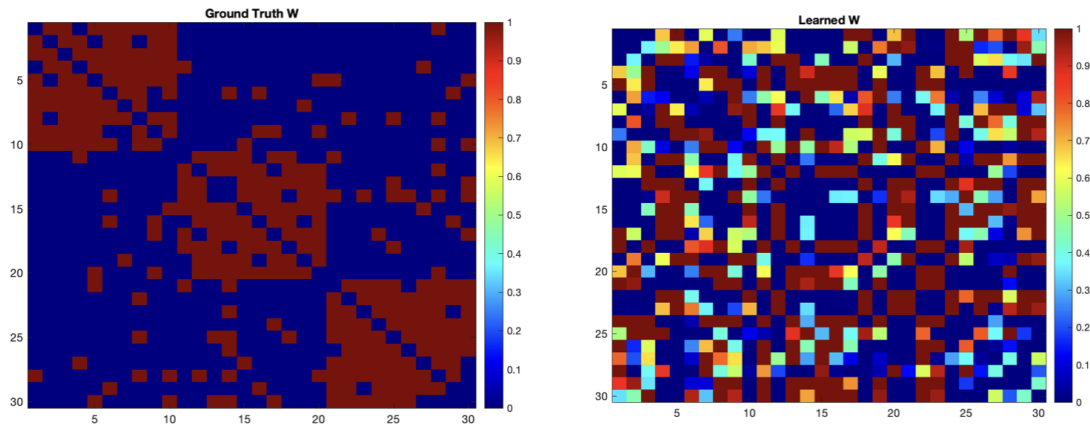
2. Formulation used by Lake and Tanenbaum

$$\begin{aligned}
 \min_{L_{\text{pre}} \succ 0, W, \sigma^2} \quad & \text{tr} \left(L_{\text{pre}} \frac{1}{p} X X^T \right) - \log |L_{\text{pre}}| + \frac{\lambda}{p} \|W\|_1, \\
 \text{s.t.} \quad & L_{\text{pre}} = \text{diag} \left(\sum_j w_{ij} \right) - W + I/\sigma^2, \\
 & w_{ii} = 0, i = 1, \dots, n, \\
 & w_{ij} \geq 0, i = 1, \dots, n; j = 1, \dots, n, \\
 & \sigma^2 > 0,
 \end{aligned} \tag{24}$$

Lake and Tanenbaum imposed the θ_{cap} constraint as a Hard one, rather than a soft which we seem to be doing in our formulation, although there the idea was to learn an Inverse Covariance Matrix itself.

Some Learning till now:





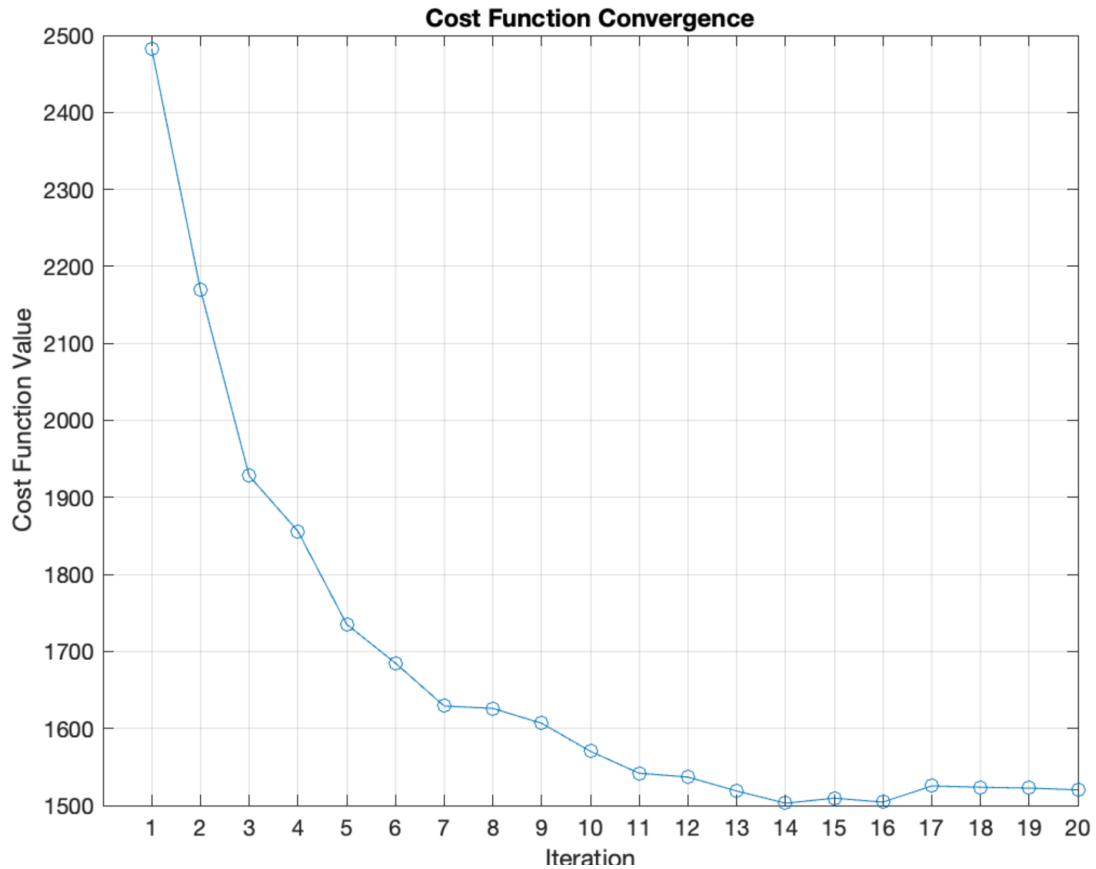
Its Clearly seen that the energy is not seem to be going down and rather its being more spiky and startpoint to endpoint energy doesnt seem to have changed at all!

Up Untill now 2 constraint we intend to impose but unable to due to Manopt:

$$1. V^* \Delta V'^* 1 = 0$$

$$2. 1^T * V = [1 \ 0 \ 0 \ \dots \ 0]$$

But the First constraint I was some how able to impose that as a constraint which had led to this decrease in energy:



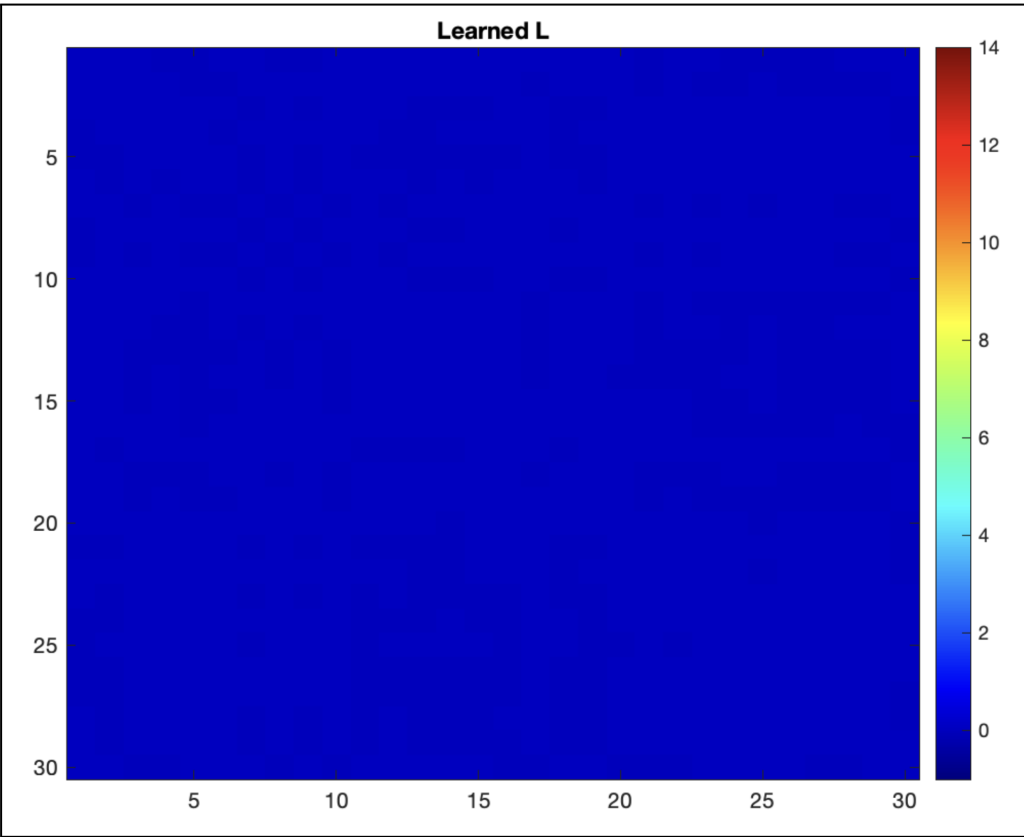
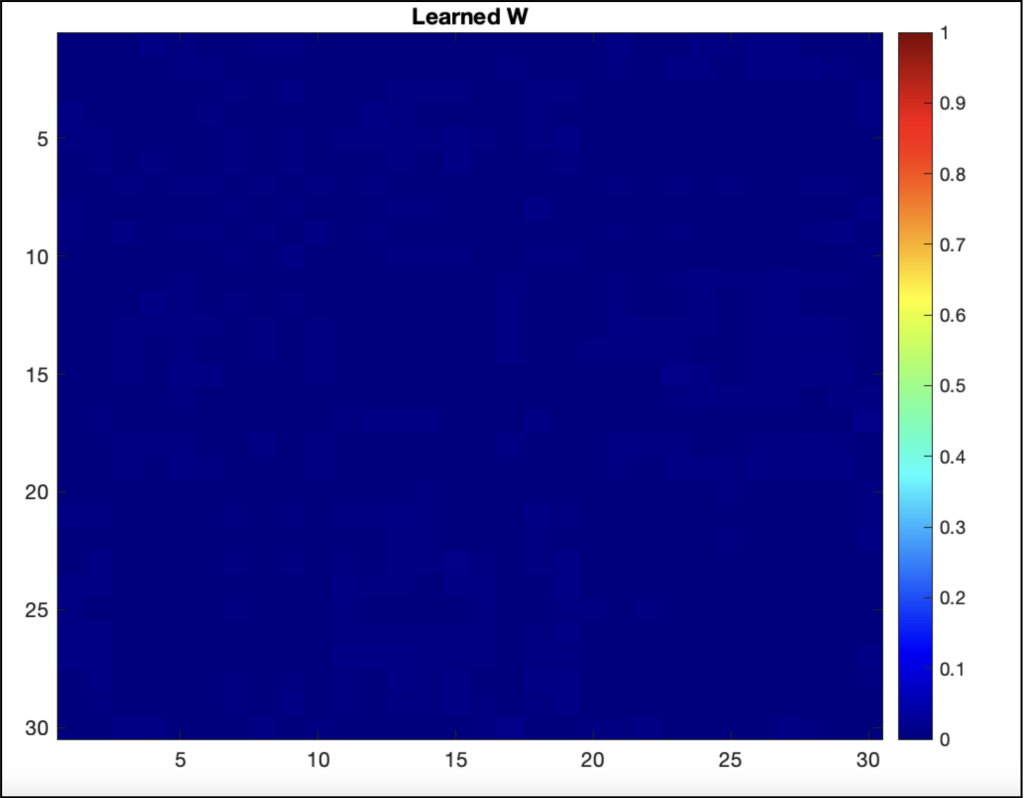
So the above plot shows a steady decline in the energy within the cost function, which is suggestive of the improvement.

However, the scale of the edge weights is getting extremely difficult to tune, as the second constraint is being imposed.

And also even if I am imposing the second constraint it doesn't seem to work as intended. Hence learned graphs almost seem to be good but the pattern is very faint.

Corresponding to it Generated plots were faint but looked like this:

Even after lot of hyper-param search this is what could be obtained.



Section 8 Imposing 2 More Constraints in the Stiefel Manifold via Manopt

Found in the Formulation that the learned V , doesn't actually have the first column as all 1's / $\sqrt{\text{size}(V,1)}$, which is pivotal for learning a Graph Laplacian L since a laplacian always has an eigenvector of this form:

$$k*[1 \ 1 \ \dots 1]^T$$

corresponding to an eigenvalue of **zero**.

So in the optimization process after performing optimizing over Stiefel manifold we found that the V 's First column should be all ones, so we used to obtain a V , then replace the first column directly with all ones/ $\sqrt{n}$ and then perform a QR decomposition, to obtain another Matrix V , which would be containing the orthogonal basis of the space spanned by the earlier columns of V . But there is a catch, when we replace the first Column with all 1's/ $\sqrt{n}$, the attained Matrix may not be Orthonormal, so performing qr on it get the orthogonalised version of the earlier matrix V seems futile.

So in the Stiefel manifold optimisation being done using Manopt, we decided to impose this as a Hard constraint:

$$\mathbf{1}^T * V = [1 \ 0 \ 0 \ \dots 0]$$

But unfortunately, due to the 'Hard Constraint' Unfriendly Nature of Manopt, this constraint cannot be incorporated within it, thus leading to immature learned V and thus immature learned L .

Have Asked on the Manopt Forum regarding this issue and this was the response:



Piyush Sawarkar

11 Oct 2023, 14:11:40 (13 hours ago) ☆ ⌨

to Manopt

Hello @Manopt Team

I had a Non-Convex optimization formulation that I was interested in solving for obtaining an Orthonormal Matrix, so this needed a search over an Orthonormal set of Matrices, now this problem being out of reach from CVX, I resorted to MANOPT.

Now following the excellent tutorial, I have defined the *problem* structure with the manifold, cost, and egrad. Also, I have given manual gradients (using matrixcalculus.com). So Manopt excelled in solving this optimization problem nicely.

Let me call my Orthonormal Matrix V and another Matrix which I do not optimize upon D (delta).

But now I was interested in imposing multiple additional HARD CONSTRAINTS like for my particular case, I was interested in :

- having such an orthonormal Matrix, whose first column is all Ones.

- $V^T \Delta V \mathbf{1} = \mathbf{0}$

So I was interested in knowing whether there is any provision that I could impose such a hard constraint and pass it to the trust region solver?

Any help would be much appreciated.



Nicolas Boumal

11 Oct 2023, 14:54:19 (12 hours ago) ☆ ⌨ ⋮

to Manopt

Hello,

I'm glad to read the tutorial was enough to get you this far.

If I understood correctly, V lives on the Stiefel manifold (`stiefelfactory(n, p)`) -- not sure if you meant to have $p = n$ (orthogonal group) or (more generally) $p \leq n$.

Currently, there is no built-in way to force the first column of V to be a constant vector (you wrote "all ones", but I'm assuming it should be "all ones / $\sqrt{n}$ " so that the first column has norm 1).

But it's definitely within the scope of Manopt to handle that constraint. Indeed, the result manifold truly is just "The set of matrices W in `Stiefel(n, p-1)` such that $W^T \mathbf{1} = \mathbf{0}$ " -- that is a manifold. The matrix $V = [\mathbf{1}/\sqrt{n}, W]$ is indeed of the required form.

So, the "proper" way to proceed would be to adapt `stiefelfactory` to allow for that extra condition. This would take a bit of work, so let me ask: did all of the above match what you had in mind, or not quite?

Now for the constraint $V^T \Delta V \mathbf{1} = \mathbf{0}$: Of course, $V^T \mathbf{1}$ is a vector whose first entry is $\sqrt{n}$, and all other entries are zero. So, $\Delta V^T \mathbf{1}$ is just [the first column of Δ] * $\sqrt{n}$. This is a known vector -- let's call it w . The constraint says: $Vw = \mathbf{0}$.

I'm not entirely sure how to handle this as a hard constraint. We might still have a manifold, but it's not immediately obvious to me. The more pragmatic approach would be to penalize $\|Vw\|$ in the cost function, with some penalty parameter (maybe in an Augmented Lagrangian style).

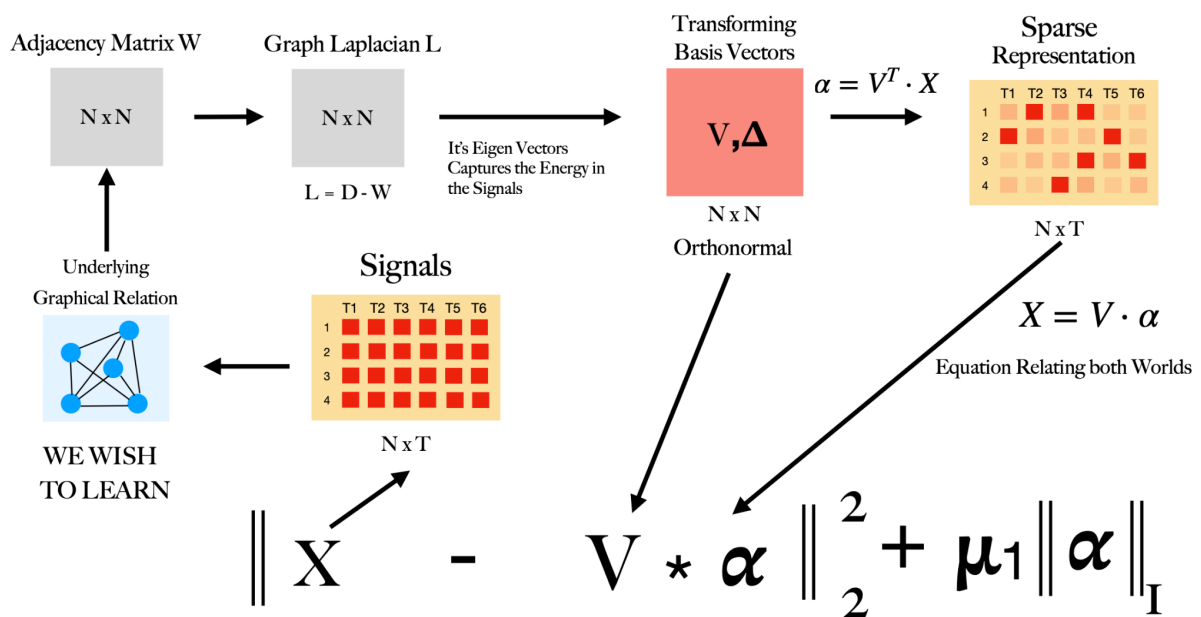
Best,
Nicolas

Link: <https://groups.google.com/g/manopttoolbox/c/YHdTRHI2ETc>

No official support for hard constraints, confirmed by the Author and their Open Source Code Base `getCost()` and `getGrad()` and `canGetCost()` `canGetGrad()` and `trustregion.m`

STEP 1

To Obtain Sparse Representation in Spectral Domain



STEP 1

$$\min_{\alpha} \|X - V * \alpha\|_2^2 + \mu_1 \|\alpha\|_I$$

Solved Using CVX

From this Step We Obtain

$$\alpha$$

STEP 2

Notion Of Sparsity

$\mathcal{L}$



Element Wise L1 norm
We Could Also Impose
L1/ρ Block Norm Regulariser

To Ensure Graph Stays Connected

$$-\mu_2 \mathbf{1}^T * \log(\text{diag}(\mathbf{V} * \Delta * \mathbf{V}^T))$$

$$+ \mu_3 \left\| \mathbf{V} * \Delta * \mathbf{V}^T \right\|_{I,I}$$

Diagram showing the relationship between the block norm regulariser and the Frobenius norm: $\left\| \mathbf{L} \right\|_{I,I}$ points to $\left\| \mathbf{V} * \Delta * \mathbf{V}^T \right\|_{I,I}$.

But there is No Data Fidelity term, so to get in that we impose the following

$$+ \mu_4 \left\| \hat{\theta} - \mathbf{V} * \Delta * \mathbf{V}^T - \frac{1}{\sigma^2} * I \right\|_2^2$$

Inverse Covariance Matrix

$$\hat{\theta} = \left(\frac{X \cdot X^T}{T-1} \right)^{-1}$$

Regulariser

$$\sigma = \min \text{EigenValue}(\hat{\theta})$$

STEP 2

$$\min_{\Delta} -\mu_2 \mathbf{1}^T * \log(\text{diag}(\mathbf{V} * \Delta * \mathbf{V}^T)) + \mu_3 \left\| \mathbf{V} * \Delta * \mathbf{V}^T \right\|_{I,I} + \mu_4 \left\| \hat{\theta} - \mathbf{V} * \Delta * \mathbf{V}^T - \frac{1}{\sigma^2} * I \right\|_2^2$$

Subject To

$$1. \mathbf{V} * \Delta * \mathbf{V}^T * \mathbf{1} = \mathbf{0}$$

$$2. \Delta \geq 0$$

Solved Using CVX

From this Step We Obtain

$$\Delta$$

STEP 3

Putting up Altogether to Obtain the V

Use α and Δ obtained in previous step to obtain V

$$\min_{\mathbf{V}} \left\| \mathbf{X} - \mathbf{V} * \alpha \right\|_2^2 - \mu_2 \mathbf{1}^T * \log(\text{diag}(\mathbf{V} * \Delta * \mathbf{V}^T)) + \mu_3 \left\| \mathbf{V} * \Delta * \mathbf{V}^T \right\|_{I,I} + \mu_4 \left\| \hat{\theta} - \mathbf{V} * \Delta * \mathbf{V}^T - \frac{1}{\sigma^2} * I \right\|_2^2$$

Subject To

$$1. \mathbf{V} * \Delta * \mathbf{V}^T * \mathbf{1} = \mathbf{0}$$

$$2. \mathbf{1}^T * \mathbf{V} = [1 \ 0 \ 0 \ 0 \dots 0]$$

Thus we compute the Laplacian from the learned V, Δ .

$$L = V * \Delta * V^T$$

$$W = \mathcal{F}(L)$$

*W_{int} <- Make Diagonal of L all Zeros
 W is Negative of W_{int}*